



Daily resilience of urban road transportation systems: Assessment, enhancement, and recovery

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ABSTRACT

As urban transportation systems face increasing pressures from rapid urbanization and recurrent daily disturbances, the ability to monitor, adapt, and recover from disruptions becomes crucial. Despite advances in resilience research, better empirical and modeling frameworks are still needed to quantify time-varying resilience under daily disturbances. This research addresses this gap by reformulating resilience into a temporal framework and introducing an indicator that captures daily fluctuations in network performance. Our experiments further reveal a Resilience Linear Gain Interval within which resilience responds linearly to link-quality improvements, enabling rapid quantification of resilience gains. This linear gain mechanism is shown to be robust across day types, demand heterogeneity regimes, and observational noise, while cross-network validation and runtime analysis support its transferability and practical efficiency. Additionally, we observe a diminishing marginal effect of resilience enhancement when simultaneously improving multiple critical links, informing how, when, and where to prioritize interventions. Beyond links, we identify resilience-critical nodes by jointly considering their structural connectivity and the demand carried by connected links, thereby uncovering bottlenecks that dominate network performance. Together, these contributions advance resilience assessment from post-disaster analyses toward a dynamic, multi-scale framework that provides actionable insights for improving urban road transportation resilience.

1. Introduction

Reliable and efficient urban road transportation systems (URTS) are recognized as the critical infrastructures of contemporary cities, underpinning routine mobility, emergency response, evacuation, and spatiotemporal allocation of critical resources [1,2]. However, with rapid urban expansion, increasing climate volatility, and surging travel demand, URTS are exposed to a wide spectrum of disturbances, ranging from recurrent daily congestion, localized incidents, and short-term facility malfunctions to large-scale disruptions caused by extreme events. Although these disturbances differ in intensity and duration, they can all undermine traffic efficiency, service continuity, and the overall functioning of urban systems. Improving the ability of URTS to withstand, adapt to, and recover from such disturbances has therefore become a central issue in transportation research and practice [3].

Resilience provides an important perspective for understanding these issues [4]. In transportation systems, resilience is generally interpreted as the capacity of a system to resist degradation, absorb disturbances, adapt to changing conditions, and recover acceptable functionality over time [2,5]. Existing studies have made substantial progress in resilience-related research, which can broadly be grouped into three closely related directions: resilience assessment, resilience enhancement, and resilience recovery [6]. However, these three directions are often studied separately. Resilience assessment typically focuses on measuring system degradation under disruption [7], resilience enhancement emphasizes pre-event strengthening or intervention [8], and resilience recovery addresses post-event restoration [9]. As a result, existing studies rarely provide a unified analytical framework that can connect resilience evaluation with enhancement and recovery decisions.

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Moreover, most current resilience studies remain oriented toward serious disruptions such as floods, earthquakes, storms, or other extreme events [10,11]. By contrast, the disturbances that dominate urban road operations on a day-to-day basis are often frequent, localized, and low-intensity, such as recurrent congestion, demand surges during peak hours, signal failures, minor crashes, and localized weather impacts [12,13]. These daily disturbances seldom cause abrupt binary failure, but instead lead to gradual and cumulative performance degradation, demand redistribution, and evolving bottleneck dependence across the network [14]. Such dynamics are difficult to capture using conventional disruption-oriented resilience approaches.

The challenge is not only how to evaluate daily resilience, but also how to translate resilience analysis into actionable intervention and recovery. At the link level, it remains unclear how incremental improvements in road operational quality translate into system-level resilience gains, which limits the development of rapid and interpretable enhancement strategies. At the node level, resilience depends not only on connectivity but also on the traffic demand carried by connected links [15], yet this joint influence is often overlooked in critical-component identification. In recovery analysis, current strategies are still largely post-disaster oriented and commonly rely on either computationally intensive optimization procedures or single-dimension indicators, offering limited support for rapid decision-making under daily disturbances [16]. Therefore, there is a need for an integrated framework that can simultaneously address daily resilience evaluation, link-quality improvement, critical node identification, and recovery prioritization.

Motivated by these considerations, we first construct a time-varying network representation of URTS as the foundation for our analysis (see Fig. 1a). Building on this, we establish a framework that connects daily resilience evaluation with link- and node-level analysis and recovery strategies. Specifically, we propose a resilience indicator that reflects the impacts of network daily performance degradation and captures the temporal dynamics of URTS daily resilience (Fig. 1b, Methods). Leveraging this indicator, we then investigate how incremental link-quality improvements affect network daily resilience. Our experiments reveal the existence of a Resilience Linear Gain Interval (RLGI) within which resilience responds linearly to link-quality changes. This finding enables rapid quantification of resilience gains and provides a practical basis for rapid optimization and operational decision-making (Fig. 1c, Methods). Moving beyond links, we identify resilience-critical nodes whose importance derives from both their structural connectivity [17] and the demand carried by their connected links [18], highlighting bottlenecks that shape overall network resilience (Fig. 1d, Methods). To translate these insights into practice, we develop recovery strategies that take link performance and node characteristics into consideration, which can be directly applied to both daily disturbances and post-disaster large-scale network failures (Fig. 1e, Methods). This integrated framework substantially improves the efficiency and scalability of resilience assessment, enhancement, and recovery in urban road networks. Furthermore, we use a case study of Nanjing, China, to demonstrate that our framework consistently aligns with travel demand fluctuations and reveals diminishing returns in link-quality improvements, highlighting the nonlinear nature of cumulative resilience enhancement. Together, these advances establish a scalable and deployable approach for evaluating, enhancing, and restoring daily resilience in urban road network systems, offering actionable guidance for traffic managers on how, when, and where to intervene. The main contributions of this study are summarized as follows:

- (1) **An integrated daily resilience framework for URTS is proposed.** The framework connects resilience evaluation, enhancement, and recovery under recurrent traffic disturbances, thereby moving beyond isolated treatment of these three research directions.

- (2) **Resilience Linear Gain Interval (RLGI).** This study theoretically and experimentally demonstrates that daily resilience gain ΔR scales linearly with link-quality improvement Δq within a theoretically feasible interval, with slope k_{ij}^t equal to the link weighted coefficient $\frac{hw_{ij}^t}{t_2 - t_1}$. To the best of our knowledge, this is the first research to offer a computational framework that explicitly quantifies the relationship between resilience enhancement and incremental link quality improvements.
- (3) **Integrated node-based resilience recovery strategy.** This study introduces a critical-node identification method that simultaneously considers node structural connectivity and the demand carried by connected links. We develop complementary resilience-recovery strategies for daily and post-disaster scenarios. For daily disruptions caused by minor disturbances, we derive a quantitative formula that enables rapid, closed-form estimation of the expected resilience gain.

The organization of the paper is as follows: Section 2 reviews related studies and identifies research gaps. Section 3 details the proposed methodology for the resilience assessment metrics, linear computational method for resilience enhancement, and recovery strategies. Section 4 introduces the study area and data sources. The results are presented in Section 5, followed by a discussion and future work in Section 6. Conclusions are given in Section 7.

2. Literature review

This section reviews existing studies from three perspectives: (1) Resilience assessment of transportation systems, (2) Resilience enhancement under disruptions, and (3) Resilience recovery strategies.

2.1. Resilience assessment of transportation systems

Resilience assessment provides the foundation for understanding how transportation systems respond to disturbances. Existing studies have developed a variety of quantitative approaches, which can generally be classified into topology-based, attribute-based, and performance-based methods [19].

Topology-based approaches rely on the static structure of the network and use topological metrics to characterize system robustness and vulnerability [20]. For example, Bruneau et al. [21] established a foundational resilience framework based on functionality loss and recovery. Akbarzadeh et al. [22] examined how travel demand and network centrality jointly affect the connectivity and resilience of urban street systems. Tang et al. [23] further adopted a topology-based perspective to evaluate the resilience of urban road networks under epidemic spreading. These studies are effective for identifying structurally important nodes and links, but they usually ignore traffic-flow dynamics and temporal operating conditions, which limit their ability to represent the gradual degradation processes commonly observed in daily urban traffic.

Attribute-based approaches extend resilience assessment by incorporating infrastructure characteristics such as capacity, redundancy, hierarchy, and component criticality. Compared with topology-based methods, they offer a more comprehensive characterization of the dynamic and heterogeneous properties of transportation components [24,25]. For example, Xu et al. [26] quantified network redundancy using travel alternative diversity and residual capacity attributes. Barker et al. [27] evaluated resilience using component-importance attributes reflecting link failure impacts.

Performance-based approaches evaluate resilience by tracking system performance throughout the disruption and recovery process. These approaches provide novel insights into how effectively transportation systems address and recover from disruptions and have gained widespread recognition. Consequently, a range of performance-based

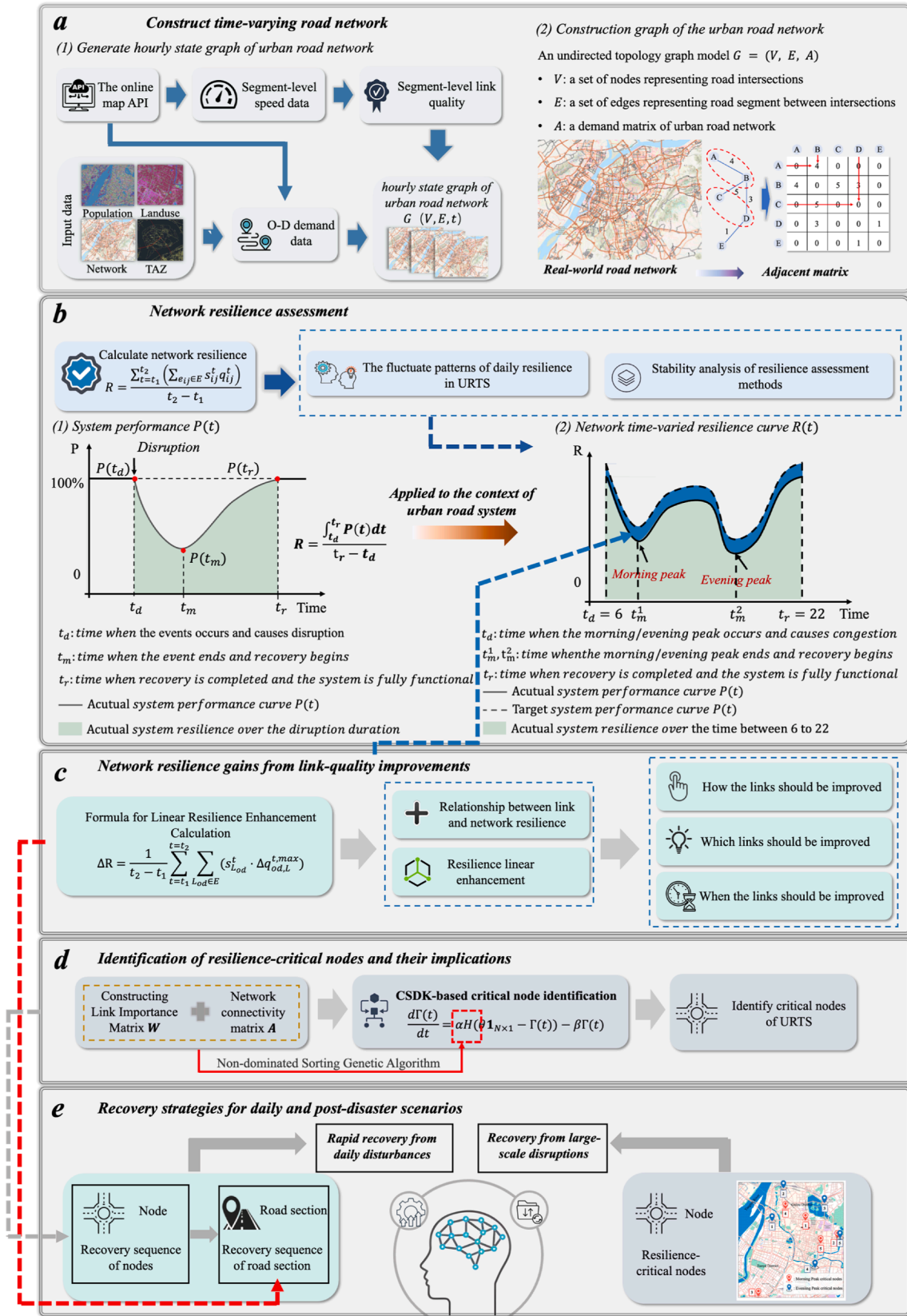


Fig. 1. Methodology framework.

indicators has been developed to characterize system performance in terms of demand-impedance, passenger volume, passenger delay, travel time, and capacity [26]. For example, Kim and Yeo [28] evaluated road network vulnerability using an MFD-based index reflecting performance loss under varying traffic states. Sun and Guan [29] measured metro vulnerability using passenger re-distribution simulations reflecting the impact of network topologies on disruptions. Li et al. [30] analyzed transit vulnerability using spatiotemporal data reflecting geographic interdependencies across subsystems. Furthermore, Murray-Tuite [31] examined transportation network resilience by comparing system optimal and user equilibrium traffic assignments. D'Lima and Medda [32] measured resilience using a mean-reverting stochastic model reflecting the rapidity of human-flow recovery from disruptive shocks. Compared with topology- and attribute-based methods, performance-based approaches are more closely tied to observed transportation outcomes and are particularly useful for linking resilience to system operation.

However, most existing performance-based resilience studies have been developed for event-driven or disaster-related disruptions. As a result, they are less suitable for representing the continuous, recurrent, and heterogeneous fluctuations that characterize the daily operation of urban road transportation systems.

2.2. Resilience enhancement under disruptions

Resilience enhancement concerns how transportation systems can be strengthened before or during disturbance progression so that performance degradation is mitigated and acceptable service can be maintained [33]. Existing enhancement strategies generally fall into two broad categories. The first seeks to improve system redundancy, for example, by expanding network capacity, adding connections, or increasing alternative route availability. Xu et al. [34] proposed a road network design tool that enhances transportation network resilience by adding new links to maximize route diversity and redundancy. Zhu et al. [1] proposed an economic evaluation framework for redundancy design in transportation networks under disruptions, showing that adding redundant links can be economically justified when resilience benefits under disruptive events are explicitly considered. Xu et al. [35] further developed complementary measures of transportation network redundancy from the perspectives of travel alternative diversity and network spare capacity, providing a systematic basis for redundancy-oriented resilience enhancement. These studies suggest that redundancy-oriented interventions can improve network adaptability by providing greater routing flexibility under disruption.

Retrofitting critical infrastructures is another important way to enhance transportation-system resilience. Rather than expanding the network by constructing new links, retrofit-based strategies improve the robustness and survivability of existing components under disruption. Zhang and Wang [36] proposed a resilience-based risk-mitigation framework for road networks, in which pre-disaster retrofit of critical bridges was explicitly compared with new bridge construction as an alternative means of improving network resilience. Wu and Chen [37] further integrated resilience modeling with pre-hazard mitigation planning and demonstrated the feasibility of proactive bridge reinforcement to support post-earthquake emergency medical response. Qu et al. [38] investigated how retrofitting critical transportation components can improve network redundancy under uncertain disruptions. From a broader pre-event planning perspective, Zhang et al. [39] proposed an economic optimization framework for resilient road pre-investment, in which critical links are upgraded to enhance transportation-system resilience under disruptive scenarios. These studies indicate that retrofitting existing infrastructures is not merely a complement to network expansion, but a core resilience-enhancement strategy in its own right.

While these strategies are valuable, they are still mainly developed in the context of major disruptions or long-term planning. Under daily

operating conditions, enhancement decisions are often more localized and incremental. In such cases, a central question is not only which component is important, but also how much system-level resilience gain can be achieved from improving the operational quality of a critical link. However, existing studies rarely provide a direct analytical relationship between marginal link improvements and system-wide resilience responses. Consequently, enhancement analyses often rely on computationally intensive traffic assignment procedures, which limit their applicability to rapid operational decision-making.

2.3. Resilience recovery strategies

Resilience depends not only on how systems resist degradation, but also on how they recover after performance loss has occurred [40]. Existing recovery studies can broadly be divided into optimization-based and indicator-based approaches. Optimization-based methods determine restoration priorities by maximizing resilience or minimizing repair time, cost, or performance loss subject to resource constraints [41,42]. Zhang et al. [43] proposed a resilience-based post-disaster recovery framework for road-bridge networks, in which restoration priorities are determined by explicitly considering damage states, traffic flow, and repair constraints. Li et al. [44] further developed a resilience-based transportation-network recovery strategy for the emergency recovery phase under uncertainty. Zhao and Zhang [45] incorporated both mobility and accessibility into transportation-infrastructure restoration optimization, broadening the objective beyond simple connectivity retention. These methods provide rigorous formulations of the recovery problem, but they often become computationally burdensome when applied to large-scale transportation networks [40].

To reduce computational complexity, many studies have adopted indicator-based recovery strategies that rank disrupted components using demand, vulnerability, centrality, or criticality measures. Sohounou and Neves [42] compared multiple link-repair sequencing strategies and showed that simple indicator-based ordering can substantially influence overall resilience recovery. Yin et al. [46] analyzed the disruption of subway nodes, edges, and lines and prioritized recovery according to their impact on passenger travel efficiency. Dong et al. [47] used traffic-speed data from real flood events to show the practical importance of repair ordering in post-flood recovery.

Despite these advances, most recovery studies remain oriented toward post-disaster restoration and are typically organized around link-based repair logic. Their applicability to daily resilience management is therefore limited, particularly when rapid intervention guidance and node-level criticality need to be considered jointly.

2.4. Research gaps

Table 1 summarizes representative studies on transportation-system resilience in terms of research target, disturbance type, and research content. Based on the above review, three major research gaps can be identified:

- (1) Mismatch between disaster-centric resilience frameworks and the gradual degradation of daily traffic operations. Although resilience strategies have been extensively developed for major catastrophic events, frameworks targeting frequent, low-intensity daily disturbances (e.g., peak-hour fluctuations, localized weather variations) remain underexplored [48,49]. The prevailing disaster-centric framing fails to capture the gradual performance degradation and cumulative efficiency erosion caused by recurrent minor events, thereby constraining the practical application of resilience in day-to-day traffic operations.
- (2) Unclear mechanisms between local link improvements and system-wide resilience. While extensive research has examined how links affect general network efficiency [50,51], it remains

Table 1
Summary of relevant research on urban transportation resilience.

Reference	Research target			Disturbance type		Research content		
	URTS	UPTS	MNTS	DD	SD	RA	RE	RR
Wang et al. [55]		✓		✓				✓
Ma et al. [56]	✓			✓		✓	✓	
Zhu et al. [1]	✓			✓				✓
Zhang et al. [6]			✓	✓		✓		
Wei and Xu [57]	✓			✓		✓		
Bi et al. [4]		✓		✓		✓		
Xu et al. [58]		✓		✓		✓		
Wang et al. [59]			✓	✓				✓
Chen et al. [60]		✓		✓		✓		✓
Ma et al. [61]			✓	✓		✓		✓
Xiao et al. [62]		✓		✓		✓		
Salvo et al. [63]	✓			✓			✓	
Liu et al. [8]			✓	✓			✓	
Qu et al. [38]	✓			✓			✓	
Liu et al. [2]		✓			✓	✓		
This paper	✓				✓	✓	✓	✓

Note: The serious disruptions considered in this paper mainly refer to those caused by natural disasters or extreme weather. Abbreviations: URTS, urban road transportation system; UPTS, urban public transportation system (rail or bus or metro system); MNTS, multi-modal transportation system; DD, daily disturbance; SD, serious disruption; RA, resilience assessment; RE, resilience enhancement; RR, resilience recovery.

unclear how incremental improvements in link performance explicitly translate into daily resilience gains. Without a direct analytical relationship, existing enhancement analyses and many intervention-oriented evaluations still rely on computationally intensive traffic assignment procedures [52,53], rendering them impractical for rapid, micro-level decision-making.

- (3) Lack of integrated link-node recovery strategies. Current recovery strategies are mostly post-disaster oriented and purely link-based. However, network resilience is jointly governed by both links and nodes. Nodes play a dual role derived from their structural connectivity and the traffic demand they carry [54]. Overlooking this nodal dimension obscures critical bottlenecks. Therefore, a framework that explicitly integrates both link- and node-level dimensions is needed for a more comprehensive understanding and management of daily resilience.

3. Methodology

This study focuses on the urban road transportation subsystem (URTS-road) under daily operational disturbances. The proposed framework is developed for resilience assessment and improvement analysis under recurrent traffic fluctuations. To maintain analytical tractability, the following assumption is adopted.

Assumption 1. (short-term demand stability). Within each analysis interval t , travelers' origins, destinations, and OD trip volumes are assumed fixed. Let $F^t = [f_{od}^t]$ denote the OD demand matrix at interval t . Although demand may vary across intervals, it is treated as constant within each hour.

This assumption is reasonable for the present study because our objective is to analyze within-day hourly resilience evolution rather than minute-level traffic oscillations. In traffic operations analysis, both 15-minute and 1-hour analysis periods are commonly used, while demand inputs are often collected or aggregated at 15-minute intervals [64]. At the same time, the present framework preserves intraday demand variability through 15-minute demand loading while conducting resilience evaluation on hourly snapshots.

The main notation is summarized in [Appendix A Table 1](#).

3.1. Network construction

The urban road network at interval t is represented as $G(t) = (V, E, A, F^t, Q^t)$, where V and E denote the sets of nodes and directed links, respectively; $A = [a_{ij}]$ is the adjacency matrix of the road network with $a_{ij} = 1$ if node i is connected to node j , and $a_{ij} = 0$ otherwise. $F^t = [f_{od}^t]$ is the OD demand matrix; and $Q^t = [q_{ij}^t]$ is the link-quality matrix describing time-varying operational conditions. To quantify the service state of each link, the link quality is defined as

$$q_{ij}^t = \frac{\min_t \tau_{ij}^t}{\tau_{ij}^t} \quad (1)$$

where τ_{ij}^t is the average travel time of link (i, j) during interval t , and $\min_t \tau_{ij}^t$ represents the minimum observed travel time of that link over the analysis horizon. By construction, $q_{ij}^t \in (0, 1]$, where a value closer to 1 indicates better operational performance and a smaller value indicates more severe congestion. Repeating this process over all intervals yields a sequence of time-dependent network snapshots $\{G(t)\}$.

3.2. Resilience assessment method

3.2.1. Path quality and optimal paths

For an OD pair (o, d) , let Ψ_{od}^t denote the set of all topologically feasible directed paths connecting o to d at interval t . For any path $\psi \in \Psi_{od}^t$, its path quality is defined as the minimum link quality along that path:

$$q_{\psi}^t := \min_{e_{ij} \in \psi} q_{ij}^t \quad (2)$$

This definition follows the bottleneck interpretation of percolation: a path becomes infeasible once the threshold exceeds the quality of its weakest link. Therefore, q_{ψ}^t characterizes the survival threshold of the path ψ . The set of optimal paths is then defined as:

$$Q_{od}^t = \arg\max_{\psi \in \Psi_{od}^t} q_{\psi}^t, \quad q_{od}^{t*} = \max_{\psi \in \Psi_{od}^t} q_{\psi}^t \quad (3)$$

All paths in Q_{od}^t share the same bottleneck value q_{od}^{t*} . Hence, as the threshold increases, the OD pair becomes disconnected when the threshold surpasses q_{od}^{t*} , because all optimal paths fail at that threshold. In this sense, q_{od}^{t*} characterizes the survival of the last feasible optimal path(s) between o and d .

3.2.2. Link weighted coefficient

For any optimal path $\psi \in Q_{od}^t$, there may exist more than one link whose quality attains the minimum path quality. We define the set of bottleneck links on ψ as:

$$\mathcal{B}(\psi, t) := \{e_{ij} \in \psi \mid q_{ij}^t = q_{\psi}^t\} \quad (4)$$

$\mathcal{B}(\psi, t)$ contains all links on path ψ whose quality equals the bottleneck value of that path. These links are the weakest links on ψ , and they determine the threshold at which the path becomes topologically infeasible. When multiple optimal paths exist, and multiple bottleneck links exist on a given optimal path, all tied candidates are treated symmetrically by splitting. Let $\mathcal{W}^t = \{(o, d) \in V \times V \mid f_{od}^t > 0\}$ be the set of active OD pairs at the interval t . The total network demand is:

$$1_{|V|}^T F^t 1_{|V|} = \sum_{(o,d) \in \mathcal{W}^t} f_{od}^t \quad (5)$$

where $1_{|V|}$ is a column vector of ones with dimension $|V|$. The product $1_{|V|}^T F^t 1_{|V|}$ sums all entries of the OD matrix F^t , and therefore represents the total network demand at interval t .

The link weighted coefficient lwc_{ij}^t of a directed link $e_{ij} \in E$ at interval t is then defined as:

$$lwc_{ij}^t = \sum_{(o,d) \in \mathcal{W}^t} \sum_{\psi \in \mathcal{Q}_{od}^t} \frac{f_{od}^t}{1_{|\mathcal{V}|}^{tr} 1_{|\mathcal{V}|}} \frac{\mathbb{I}\{e \in \mathcal{B}(\psi, t)\}}{|Q_{od}^t| |\mathcal{B}(\psi, t)|} \quad (6)$$

$\mathbb{I}\{e \in \mathcal{B}(\psi, t)\}$ is the indicator function, which equals 1 if the link e belongs to the bottleneck set $\mathcal{B}(\psi, t)$, and 0 otherwise. This two-level equal-splitting rule first allocates each OD demand equally across its optimal paths and then equally across the tied bottleneck links on each optimal path. It also ensures that:

$$lwc_{ij}^t \geq 0, \quad \sum_{e \in E} lwc_{ij}^t = 1 \quad (7)$$

If a link is never identified as a bottleneck link on any optimal path for any active OD pair with positive demand, then $lwc_{ij}^t = 0$.

In implementation, the optimal bottleneck value $q_{od}^{t,*}$ and its associated bottleneck-link candidates are obtained using a modified Dijkstra-type algorithm that maximizes the minimum link quality, thereby avoiding explicit enumeration of the full path set Ψ_{od}^t .

3.2.3. Resilience assessment metric

To quantify the system's ability to withstand and recover from daily disturbances, a time-dependent resilience metric is defined by aggregating link quality weighted by its bottleneck importance. Following the resilience-triangle concept [65–67], resilience over a continuous time window $[t_1, t_2]$ is expressed as:

$$R_{[t_1, t_2]} = \frac{\int_{t_1}^{t_2-1} \sum_{e_{ij} \in E} lwc_{ij}^t q_{ij}^t dt}{t_2 - t_1} \quad (8)$$

Since the present analysis is conducted at a discrete temporal resolution, the above expression is approximated by its discrete-time counterpart:

$$R_{[t_1, t_2]} = \frac{\sum_{t=t_1}^{t_2-1} \left(\sum_{e_{ij} \in E} lwc_{ij}^t q_{ij}^t \right)}{t_2 - t_1} \quad (9)$$

$$w_{od}^t = \begin{cases} lwc_{ij}^t, & \text{if the link } (i,j) \text{ is the limiting link from node } o \text{ to node } d \text{ during hour } t, \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

Eq. (9) measures the average network service level over the selected time window, where link performance is weighted by its contribution to OD connectivity under the bottleneck-path structure.

3.3. Network resilience gains from link-quality improvements

In this study, improving q_{ij}^t corresponds to reducing the average travel time on the link (i,j) , for example, through capacity enhancement, congestion mitigation, or operational management. Importantly, not every marginal increase in link quality changes the optimal path structure. Based on this observation, we define a specific improvement regime, termed the Resilience Linear Gain Interval (RLGI).

Within the RLGI, improving a bottleneck link changes only its local quality level without altering the optimal path set $\mathcal{Q}_{od}^t$ or their bottleneck-link sets $\mathcal{B}(\psi, t)$. As a result, the OD allocation logic remains unchanged, and lwc_{ij}^t remains invariant. Under this condition, the network resilience response is linear with respect to the quality increment. Once the improvement exceeds the upper structural bound q_{safe}^t , the optimal path structure may change, and the network state must be

re-evaluated accordingly. (The details of the RLGI boundaries and the algorithmic exploitation are illustrated in Appendix B and C, Algorithms 1–2).

Therefore, if the link-quality improvements Δq_{ij}^t remain strictly within the RLGI, the resilience gain over the time window $[t_1, t_2]$ is:

$$\Delta R_{[t_1, t_2]} = \frac{1}{t_2 - t_1} \sum_{t=t_1}^{t_2-1} \left(\sum_{e_{ij} \in E} lwc_{ij}^t \cdot \Delta q_{ij}^t \right) \quad (10)$$

Eq. (10) provides an analytical evaluation of resilience gains without repeatedly re-solving traffic assignment for each marginal intervention, provided that the improvement remains within the RLGI.

As long as the intervention remains within the RLGI, the optimal path structure and bottleneck-link sets do not change and lwc_{ij}^t remains fixed. Therefore, Eq. (10) can be used directly to evaluate resilience gains. UE reassignment is required only when an intervention exceeds the RLGI boundary, because the routing structure may then change, and the corresponding flow pattern must be updated.

3.4. CSDK-based identification of resilience-critical nodes

While link-level analysis reveals where resilience degradation is concentrated, practical traffic management is often implemented at the node level, such as at intersections, control points, or transfer hubs. A node may become critical not only because of its topological position, but also because it connects resilience-sensitive or high-demand links. To capture this dual role, Fig. 2 presents the proposed CSDK-based framework for identifying resilience-critical nodes.

Specifically, a time-varying traffic weight matrix H^t is constructed by combining the link-importance matrix W^t , derived from the link weighted coefficients lwc_{ij}^t , and the static connectivity matrix A :

$$H^t = a * W^t + b * A, \quad a + b = 1 \quad (11)$$

Here, $W^t = [w_{od}^t]$ denotes the criticality-based link weighted coefficient matrix,

which reflects the dynamic contribution of resilience-critical links. The matrix $A = [a_{od}]$ denotes the static connectivity matrix,

$$a_{od} = \begin{cases} 1, & \text{if node } d \text{ is reachable from node } o \text{ in the static network,} \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

which captures the underlying static reachability structure of the network. Based on this weighted network, node criticality is quantified through a Domirank-based competitive dynamics model, and the stationary dominance score Γ is used to rank resilience-critical nodes. In computation, the score is obtained recursively as

$$\Gamma(t + dt) = \Gamma(t) + (\sigma H^t (\theta 1_{N \times 1} - \Gamma(t)) - \Gamma(t)) dt \quad (14)$$

where σ controls the balance between local dominance and global network effects. In this way, the proposed framework translates link-level bottleneck information into node-level criticality while preserving both structural and demand-related influences.

The parameters a and b balance the relative influence of demand-

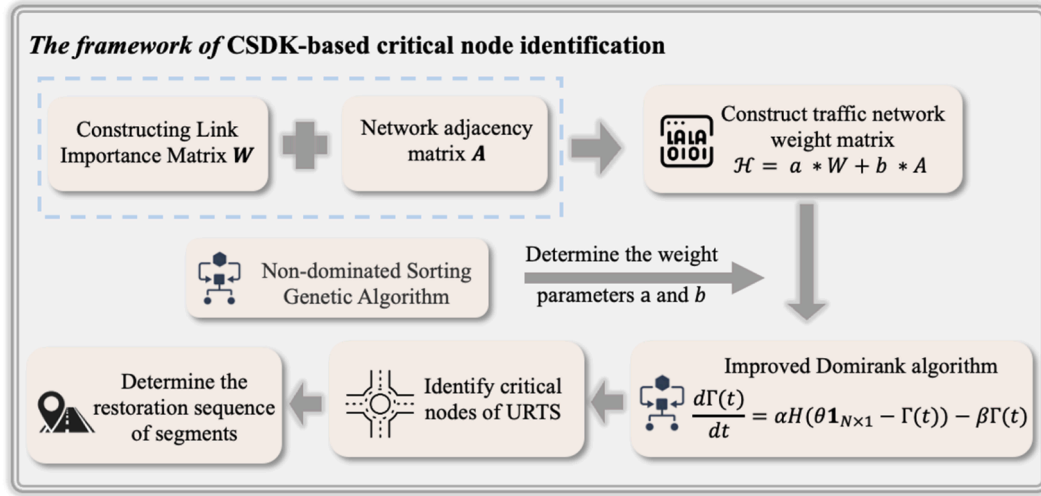


Fig. 2. The framework of CSDK-based critical node identification.

sensitive link criticality and static network connectivity in the traffic weight matrix H^t . In this study, their values are determined in a data-driven manner for each hour by jointly evaluating two criteria, namely the unaffected-demand area under the curve (UD_AUC) and the largest-connected-component area under the curve (LCC_AUC); the final parameters a and b are selected from the Pareto-optimal set using the knee-point rule. The parameter σ , which controls the balance between local node dominance and global network effects in the Domirank dynamics, is determined by searching its feasible interval for the value that yields the best evaluation performance. Thus, both the weighting parameters (a, b) and the dominance parameter σ are adaptively calibrated rather than fixed a priori. Detailed derivations of the CSDK dynamics,

together with the definitions of UD_AUC and LCC_AUC , as well as the calibration procedures for (a, b) and σ , are provided in [Appendix D](#).

3.5. Node-based resilience recovery framework

To translate critical-node identification into actionable decision support, a node-based resilience recovery framework is developed. Network disruptions are simulated by sequentially removing and restoring individual nodes. When a node is removed, all of its incident links are assumed to fail simultaneously, preventing traffic flow through the affected local structure. Recovery is modeled by reactivating the failed node together with its incident links, thereby restoring

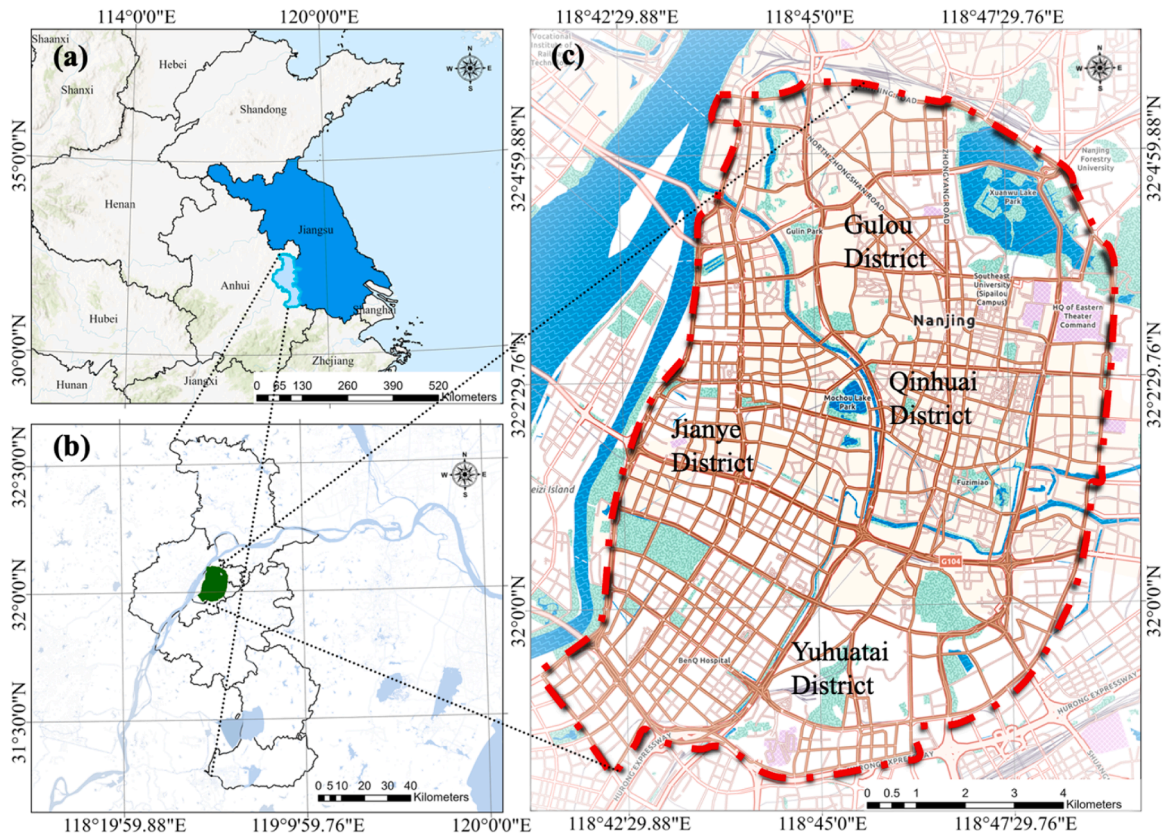


Fig. 3. Location of the study area.

connectivity and allowing traffic to resume.

For comparison, five recovery strategies are considered: degree-based recovery, betweenness-based recovery, PageRank-based recovery, demand-based recovery, and the proposed CSDK-based recovery strategy [63,68,69]. For all strategies, disrupted nodes are restored according to their corresponding priority rankings, and recovery performance is evaluated using the network-wide resilience metric R defined in Eq. (9). Two recovery scenarios are considered.

(1) Recovery under daily disturbances

This scenario represents short-horizon disturbances such as peak-hour crashes, congestion surges, and weather-induced localized gridlock, where traffic managers require rapid and interpretable recovery guidance. Under this setting, travelers' origins, destinations, and trip volumes are assumed to remain unchanged, and once a critical node fails, the associated trips are treated as interrupted rather than immediately rerouted. Under these quasi-static assumptions, the corresponding link weighted coefficients remain unchanged, allowing Eq. (10) to be directly used to estimate the resilience improvement brought by restoring critical nodes.

(2) Recovery under post-disaster disruptions

This scenario addresses large-scale network outages caused by natural disasters or extreme weather, where restoration unfolds over a longer time horizon. In contrast to the daily-disturbance scenario, no quasi-static assumptions are imposed. Recovery is therefore evaluated dynamically as failed nodes are progressively restored and network resilience is recalculated throughout the restoration process.

4. Region and data preparation for case studies

The case study is conducted in the core urban area of Nanjing, covering the Gulou, Jianye, Xuanwu, and Yuhuatai districts (Fig. 3). To support the proposed resilience framework, multi-source datasets are integrated to construct the road network, estimate time-dependent demand, derive link-level operating states, and assess the fidelity of the simulated traffic environment.

4.1. Data sources

The data used in this study were obtained mainly from the Nanjing

Table 2
Data sources for regional road network construction and demand estimation.

Category	Data	Sources	Resolution
The URTS topology data	Road network topology	Open Street Map (https://www.openstreetmap.org/)	Vector
Traffic demand data	Population density data	2024 Nanjing Statistical Yearbook	100 m
	Age structure	2024 Nanjing Statistical Yearbook	100 m
	DEM	Geospatial Data Cloud (https://www.gscloud.cn/)	30 m
	Administrative boundary	Geospatial Data Cloud (https://www.gscloud.cn/)	Vector
	Traffic analysis zone	Open Street Map (https://www.openstreetmap.org/)	Vector
Link average speed data:	Travel time between two TAZs	The Gaode Map Route Planning API (https://lbs.amap.com/api/webservice/guide/api/direction)	16 days
	Road segments operation status data	The Gaode Map Traffic Operation States API (https://lbs.amap.com/api/webservice/guide/api-advanced/traffic-situation-inquiry)	16 days

Statistical Yearbook, OpenStreetMap (OSM), Geospatial Data Cloud, and the Gaode Map platform. As summarized in Table 2, these datasets provide the road-network topology, zonal demographic attributes, topographic and administrative information, time-dependent travel impedance, and observed traffic operation states required for road-network construction, demand estimation, traffic-state generation, and model validation in the case study.

4.2. Preparation procedure

The study network is constructed by cross-filtering the OSM road topology with traffic operation records obtained from the Gaode Map API. Only road segments with valid traffic-state observations are retained, and isolated links outside the main connected component are removed. This yields the largest connected road network component used in the subsequent resilience analysis.

To ensure consistency across the integrated datasets, all inputs are aligned to the same 16-day study period. Dynamic OD matrices $F^t = [f_{od}^t]$ are first generated at 15-minute intervals to capture intraday demand variation. Population density and age-structure data are mapped to Traffic Analysis Zones (TAZs) to estimate trip generation and attraction, and inter-zonal trips are then distributed using a gravity model calibrated with time-dependent travel times from the Gaode Route Planning API.

The resulting 15-minute OD matrices are loaded into a simulation-based dynamic traffic assignment model [70], in which time-dependent traffic states are generated through iterative network loading and route updating over the study horizon. Based on the simulation outputs, hourly link travel times τ_{ij}^t are extracted as the operational states for resilience analysis and are subsequently converted into the link-quality measure q_{ij}^t defined in Eq. (1). These q_{ij}^t values then serve as the direct input for bottleneck-path identification, calculation of the link weighted coefficient lwc_{ij}^t , and resilience evaluation.

In this way, the framework preserves short-term demand variability through 15-minute loading while conducting resilience evaluation on an hourly snapshot basis, providing a practical balance between temporal fidelity and analytical tractability.

Finally, the simulated traffic environment is validated against observed traffic operation data from the Gaode Map API. Because the Gaode platform provides denser and more reliable observations for major arterial roads, the validation is conducted on the principal road segments within the cross-filtered network. The comparison yields a R^2 value of 0.8126, indicating a strong goodness-of-fit that comfortably aligns with the recognized benchmark standards (65–90 %) in contemporary macroscopic traffic simulations [71,68]. These prepared inputs provide the empirical basis for the subsequent resilience assessment, enhancement analysis, and recovery evaluation.

5. Results

The empirical analysis proceeds in a stepwise manner. A simulation-based dynamic traffic assignment is first performed over the study horizon to establish the baseline traffic states. Time-dependent network snapshots are then extracted to obtain the equilibrium link travel times τ_{ij}^t , from which the link qualities q_{ij}^t are computed using Eq. (1). Based on these time-dependent snapshots, the optimal path sets q_{od}^t and the link weighted coefficients lwc_{ij}^t are derived, enabling the identification of resilience-critical links across time periods. These link-level results are then used to evaluate network resilience and analytically quantify the resilience gains associated with incremental link-quality improvements through the proposed RLGI formulation. To validate this analytical relationship, we further conduct a numerical verification by incrementally modifying the quality of selected critical links, re-solving the UE assignment, and comparing the resulting resilience improvements with

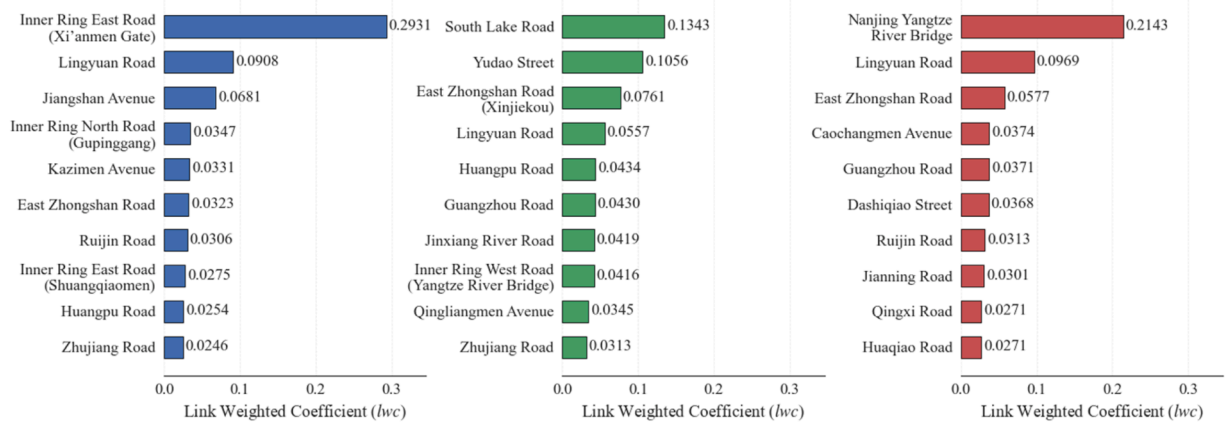


Fig. 4. Spatiotemporal distribution of the top-8 critical links across different periods on May 29.

the formula-based predictions. Building on the identified link-level bottlenecks, node-level criticality is further assessed through the CSDK-based framework, and the resulting critical nodes are finally used to analyze recovery strategies under both daily disturbance and post-disaster scenarios. Following this workflow, the subsequent subsections report the empirical findings on critical-link identification, resilience enhancement, node criticality, and recovery performance.

5.1. Spatiotemporal identification of critical links

The hourly ranking of lwc_{ij}^t reveals clear spatiotemporal variation in network vulnerability (Fig. 4). During the morning peak, critical links are concentrated along major commuting corridors leading to the urban core; at noon, vulnerability shifts toward locally important roads near dense commercial and medical areas; and during the evening peak, cross-river corridors become dominant, with the Nanjing Yangtze River Bridge ranked first. These patterns indicate that the proposed lwc_{ij}^t captures temporally shifting bottleneck dependence under realistic daily travel rhythms in Nanjing.

Importantly, lwc_{ij}^t reflects structural vulnerability rather than traffic volume alone. For instance, Lingyuan Road remains highly ranked across multiple periods, not because it carries the largest traffic volume, but because its limited capacity makes it highly sensitive to modest flow increases and thus critical to local connectivity. The ranking pattern of lwc_{ij}^t further indicates that network resilience is disproportionately governed by a small number of bottleneck links.

Comparison with the Amap Traffic Analysis Report further supports the practical relevance of the identified links. As shown in Table 3, 8 out

of 10 representative empirical congested roads are captured by the proposed method, corresponding to an overlap rate of 80 %. The two unmatched empirical roads, Hongshan Road and Puzhen Avenue, do not contradict the validity of the proposed metric. The lwc_{ij}^t indicator is designed to identify links with high structural vulnerability under the bottleneck-path framework, rather than all roads with observed congestion. Some roads may experience persistent congestion because of high local demand or corridor-specific operational frictions, while contributing less to bottleneck-dependent OD connectivity at the network level. Conversely, the proposed framework can identify links that may not exhibit the highest absolute delay but still play a disproportionate role in sustaining overall network resilience. Therefore, the identified critical links should be understood as a structurally informed complement to empirical congestion reports rather than a simple replication of them.

5.2. Network resilience gains from link-quality improvements

One of the main objectives of this study is to examine how resilience enhancement relates to link-quality improvements. As shown in Fig. 5 and Fig. 6, the proposed resilience metric R increases with link quality q for the single most critical link across the representative weekday and weekend periods. Within the theoretical interval $[0, q_{safe} - q_{od,L}^{max}]$, the fitted slope k_{ij}^t coincides with the theoretical value $\frac{lwc_{ij}^t}{(t_2 - t_1)}$ (see Table 4), confirming a strictly proportional relationship between resilience gain and link-quality improvement. Once the intervention exceeds this interval, the response becomes nonlinear: resilience continues to increase, but at a declining and increasingly irregular rate, reflecting the emergence of higher-order network interactions.

By contrast, the average shortest-path speed exhibits a smoother and nearly monotonic increase with q across the same periods. This suggests that, although speed captures overall operational efficiency, it is less sensitive to bottleneck-dependent structural vulnerability. The proposed resilience metric therefore provides complementary information by distinguishing the proportional improvement regime from the nonlinear deviation regime.

The lower-row panels (Fig. 5(d)-(f) and Fig. 6(d)-(f)) show the resilience gain rates $(\frac{\Delta R_{od,L}}{\Delta q_{od,L}} * 100\%)$. A clear two-stage pattern emerges.

Within the theoretical interval, the gain rates remain stable at 29.31 %, 13.43 %, and 21.43 % for the three representative weekday periods, and at 13.56 %, 13.20 %, and 10.78 % for the corresponding weekend periods, indicating stronger resilience gains on weekdays. Once the feasible interval is exceeded, marginal benefits decline sharply. For example, during the weekday evening peak, the gain rate drops from 21.43 % to 11.99 %, demonstrating the law of diminishing returns. Further link-quality improvements yield smaller and less predictable

Table 3

Cross-validation of the identified critical links.

Empirical Congested Link	Identified by lwc	Matching Context
Guangzhou Road (Zhongshan Rd to Hujia Rd)	Yes	Noon (Rank 6), Evening (Rank 5)
Nanjing Yangtze River Bridge	Yes	Evening (Rank 1)
Inner Ring East Road (Xinzhuang to Chenguang)	Yes	Morning (Rank 1, 8)
Inner Ring North Road (Gupinggang area)	Yes	Morning (Rank 4)
Daqiao North Road	Yes	Evening (Rank 8, corridor sync)
Jiangshan Avenue (near G312)	Yes	Morning (Rank 3)
Hongshan Road	No	None
Puzhen Avenue	No	None
Zhujiang Road	Yes	Morning (Rank 10), Noon (Rank 9)
East Zhongshan Road	Yes	Morning (Rank 6), Noon (Rank 3), Evening (Rank 3)

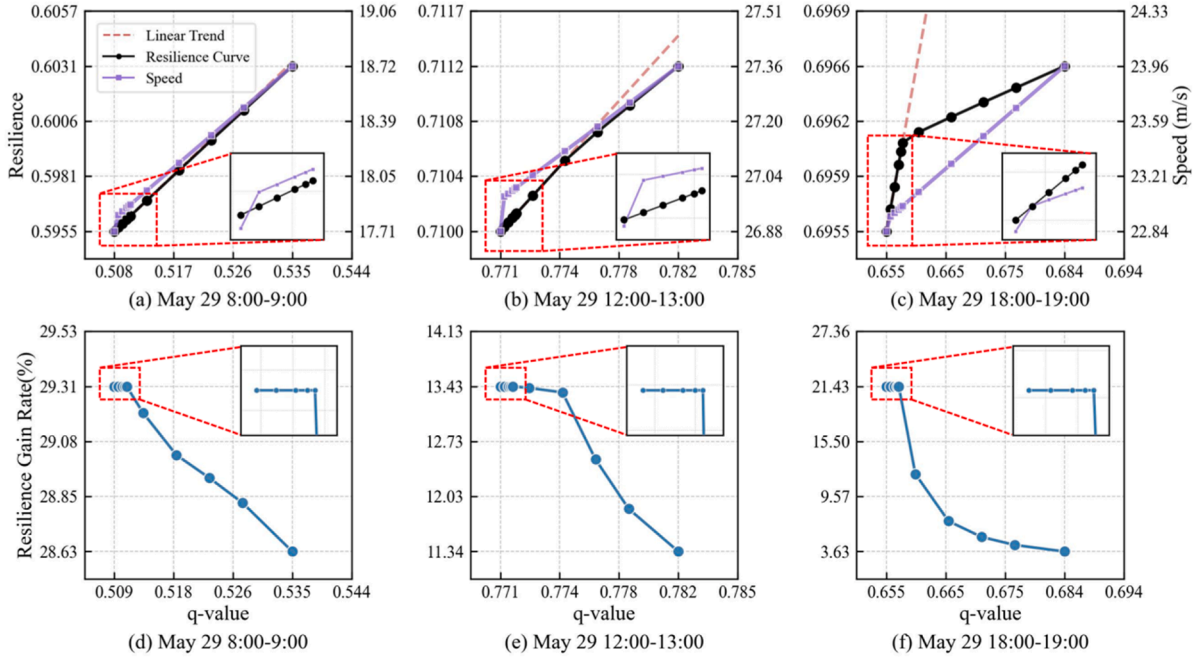


Fig. 5. Resilience, average shortest-path speed and resilience gain with link-quality improvements of the single most critical link across different time periods on May 29, 2025.

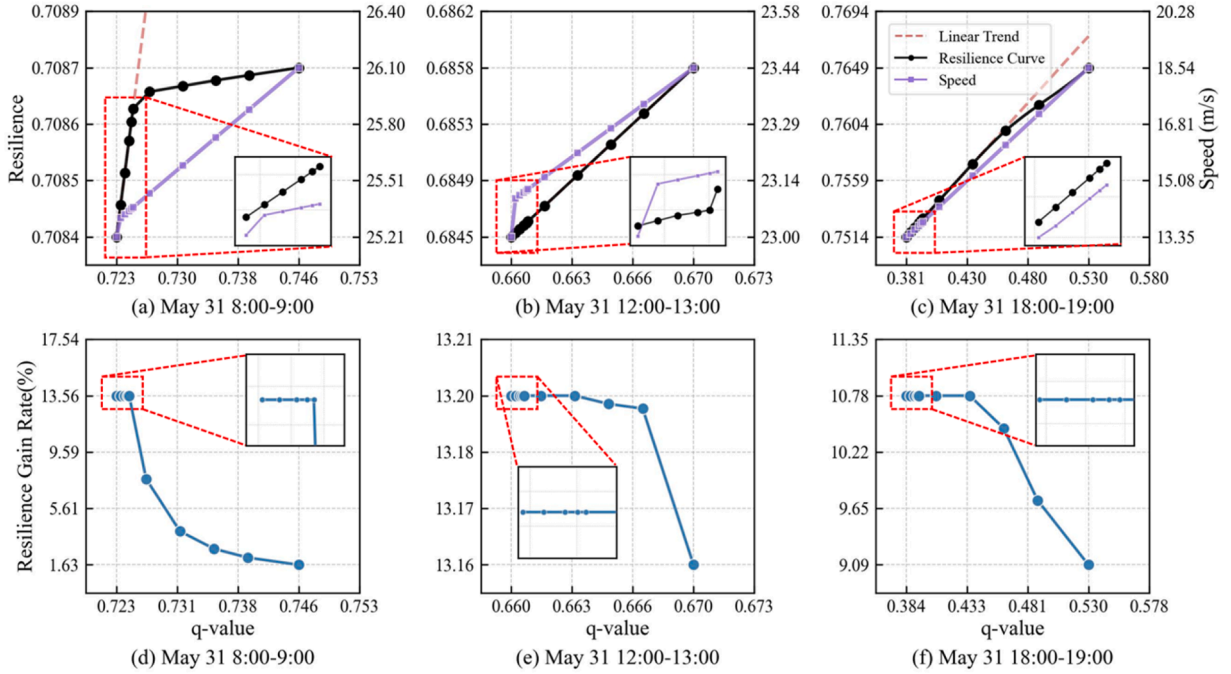


Fig. 6. Resilience, average shortest-path speed and resilience gain with link quality improvements of the single most critical link across different time periods on May 31, 2025.

Table 4

The link weighted coefficient lwc_{ij}^t for the single most critical link during three typical periods on May 29 and May 31.

lwc_{ij}^t	Morning peak (8:00–9:00)	Midday (12:00–13:00)	Evening peak (18:00–19:00)
May 29	0.29307	0.13429	0.214299
May 31	0.135647	0.131957	0.107803

resilience gains.

To extend the single-link analysis, we further examine the hourly resilience gain rate of the most critical link and compare the resilience gains associated with the top five critical links (Fig. 7). On May 29 (weekday), the gain rate peaks at 29.31 % during the morning period (08:00–09:00) and 21.43 % during the evening period (18:00–19:00), reflecting strong commuter-driven congestion effects, whereas the midday period (12:00–13:00) shows a lower but still stable gain of 13.43 %. On May 31 (weekend), the gain rates fluctuate less overall, indicating

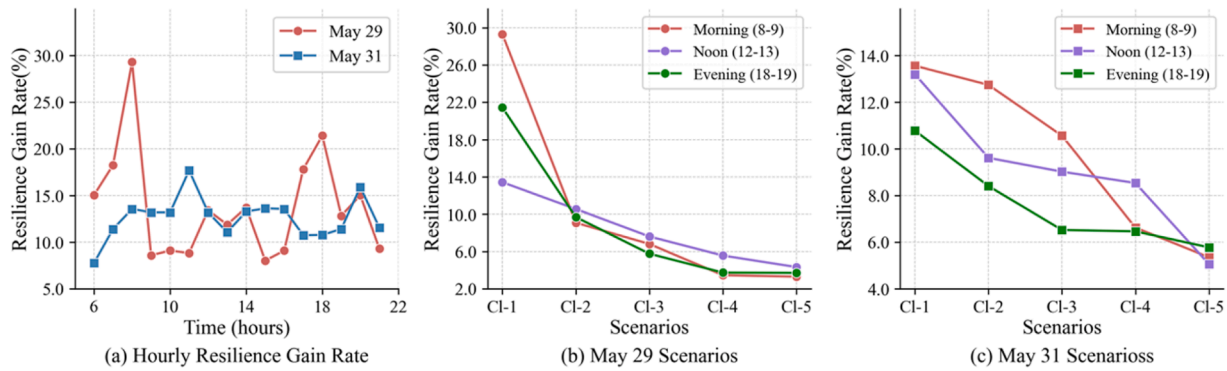


Fig. 7. Resilience gain rates analysis.

more stable and leisure-oriented traffic conditions.

For the top five links, weekday results show a clear decline in resilience gains as link criticality decreases. Replacing the most critical link (CI-1) with CI-2 reduces the gains from 29.31 % to 9.08 % in the morning, from 13.43 % to 10.56 % at midday, and from 21.43 % to 9.69 % in the evening. This sharp drop indicates that weekday peak resilience depends strongly on a single dominant corridor. By contrast, weekend gains are consistently lower and less variable, with flatter curves and stronger diminishing returns, indicating that more evenly distributed, leisure-driven traffic yields smaller marginal benefits from link-quality improvements.

Overall, these results show that resilience enhancement is both time-dependent and link-specific. The proposed RLGI provides a reliable range within which resilience gains remain strictly proportional to link-quality improvement, whereas the observed nonlinear deviation beyond the interval highlights the importance of targeted rather than uniform upgrades. Weekday congestion periods are particularly sensitive to improvements in the most critical links and therefore yield disproportionately large resilience gains, whereas weekend traffic patterns display flatter responses and faster saturation. More broadly, the proposed resilience metric R is broadly consistent with conventional efficiency indicators such as speed, while additionally capturing structural bottlenecks and nonlinear system responses that are not directly revealed by speed alone.

5.3. Robustness, transferability, and computational applicability of the proposed framework

The preceding results have shown that the proposed framework can identify resilience-critical links, characterize the linear gain regime within the RLGI, and reveal time-dependent enhancement patterns under representative weekday and weekend conditions. To move beyond these illustrative cases, this subsection further examines whether the observed patterns remain stable across the full sample, across different demand heterogeneity regimes, and under observational noise, and whether the within-interval linear validity of the framework transfers to a benchmark network with a different topology and scale. Finally, the computational efficiency of the proposed analytical

framework is evaluated to assess its practical suitability for rapid operational use.

5.3.1. Temporal stability across day types and demand heterogeneity

While the preceding analysis has demonstrated the linear gain mechanism under representative weekday and weekend periods, it remains necessary to examine whether this pattern is stable beyond a few illustrative cases. To this end, we extend the analysis to the full 16-day sample and further separate the observations into weekday and weekend subsets. As reported in Table 5, the median linearity error remains on the order of 10^{-14} for all categories, while the zero-error ratio consistently exceeds 91 %. These results indicate that the observed RLGI linearity is stable across day types rather than being driven by a limited subset of observations.

Beyond day-type variation, we further examine whether changes in demand concentration alter the stability of the proposed framework. All 256 snapshots were grouped into high-, mid-, and low-heterogeneity regimes according to the Gini coefficient. As shown in Table 6, the median linearity error remains at the 10^{-14} scale across all groups, and the zero-error ratio stays above 91%. Although the exact top-ranked bottleneck changes frequently across snapshots, the rank-1 dominance remains consistently above 1.38, indicating that the prioritization logic is structurally stable under different demand concentration patterns.

Together, these results suggest that the proposed framework remains stable across both day-type variation and substantial changes in demand concentration, thereby strengthening the credibility of the subsequent critical-component identification and recovery analysis under different traffic-demand regimes.

5.3.2. Robustness against observational noise

Because the proposed framework derives link quality, valid linear intervals, and critical-link rankings from observed or simulated traffic states, its sensitivity to observational perturbations is further examined. Multiplicative Gaussian noise with standard deviation $\zeta \in \{0.01, 0.03, 0.05, 0.10\}$ is introduced, and the effects on RLGI length, within-interval linearity, and ranking consistency are evaluated.

The results show that observational noise does not eliminate the existence of the RLGI, but primarily affects its effective length. As the noise level increases, the total RLGI length exhibits a systematic contraction, with the mean change decreasing from approximately -0.0007 at $\zeta = 0.01$ to -0.0289 at $\zeta = 0.10$. This indicates that stronger noise narrows the range within which strict linearity can be safely guaranteed, rather than invalidating the linear-gain mechanism itself. Importantly, the linear relationship remains numerically exact within the surviving interval. As summarized in Table 7, across all tested noise levels, the fitted within-interval R^2 remains equal to 1.0, and the slope errors remain negligible, with the 95th-percentile error staying on the order of 10^{-10} . This confirms that, once the intervention remains within the noise-perturbed RLGI, the proportional relationship between

Table 5
Robustness of the RLGI linearity across the full sample and day types.

Category	Day Count	Total Snapshots	Sample Size	Median Error	Zero-Error Ratio(%)
Full sample	16	256	2560	3.04×10^{-14}	91.9118
Weekday	11	176	1760	3.16×10^{-14}	91.7045
Weekend	5	80	800	2.94×10^{-14}	92.2917

Table 6

Robustness of RLGI and rank-1 link dynamics under different demand heterogeneity regimes.

Category	Snapshots	Gini (Median)	Median Error	Zero-Error Ratio(%)	Rank-1 Switch Rate(%)	Rank-1 Dominance
High Heterogeneity	84	0.167	2.76×10^{-14}	92.38	91.67	1.57
Mid Heterogeneity	87	0.151	4.07×10^{-14}	91.95	88.51	1.38
Low Heterogeneity	85	0.107	2.75×10^{-14}	91.41	94.12	1.57

Table 7

Robustness of RLGI length and within-interval linearity under observational noise.

Noise level ζ	Mean RLGI length change	Mean fitted (R^2) within RLGI	Slope error (p_{95}) within RLGI	Mean relative max residual within RLGI
0.01	-0.000695	1.0000	2.08×10^{-10}	9.25×10^{-11}
0.03	-0.005101	1.0000	2.13×10^{-10}	1.41×10^{-10}
0.05	-0.012873	1.0000	1.62×10^{-10}	2.39×10^{-10}
0.10	-0.028909	1.0000	1.78×10^{-10}	1.68×10^{-10}

Note: Negative values of RLGI length change indicate contraction of the valid linear interval relative to the baseline case.

Table 8

Robustness of critical-link ranking under observational noise.

Noise level	Coverage within RLGI (%)	Top-10 overlap (%)	Kendall's τ
0.01	100	77.000	0.9228
0.03	100	59.125	0.8936
0.05	100	51.500	0.8733
0.10	100	40.688	0.8143

resilience gain and link-quality improvement still holds strictly.

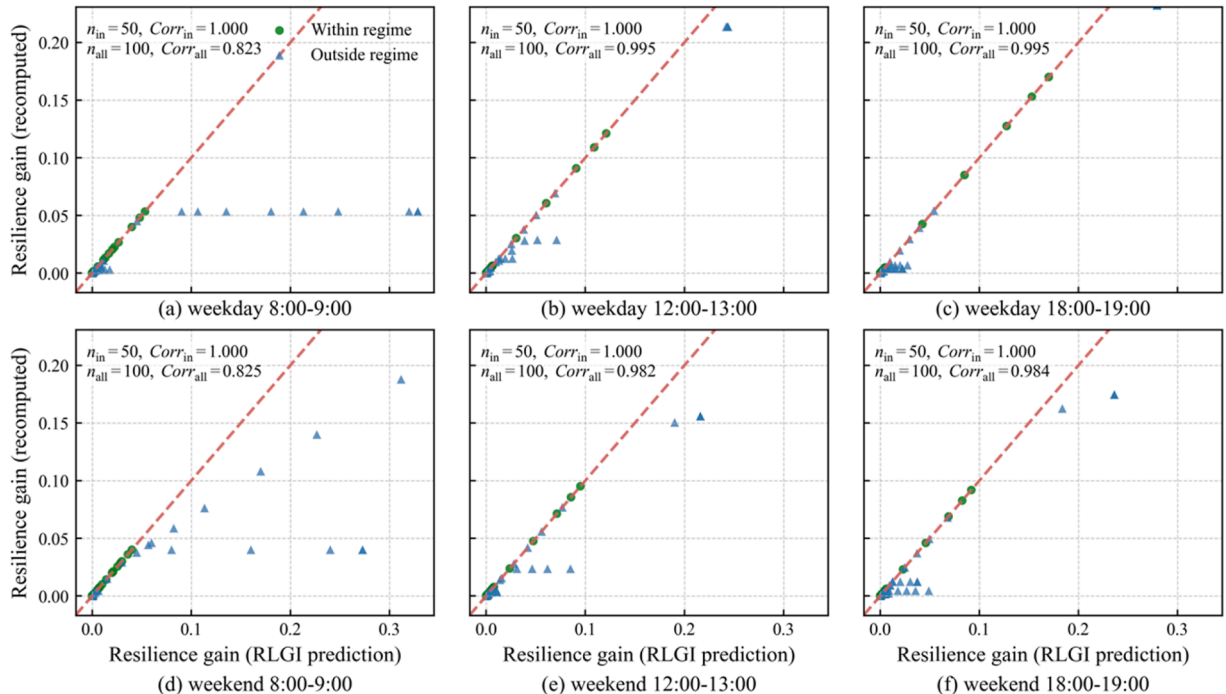
The ranking results are more sensitive to noise than the within-interval slope itself, but they remain broadly stable in an overall sense. To quantify the agreement between the baseline and noise-perturbed critical-link rankings, Kendall's τ is adopted as a nonparametric rank-correlation measure, consistent with its original formulation for comparing ranked orderings [69]. As reported in Table 8, the

valid RLGI remains identifiable under all tested noise levels, with coverage within the RLGI remaining at 100 %. By contrast, the exact composition of the highest-ranked links becomes progressively more sensitive to noise: as ζ increases from 0.01 to 0.10, the Top-10 overlap declines from 77.0 % to 40.7 %, whereas Kendall's τ decreases more moderately from 0.923 to 0.814. This suggests that observational noise mainly perturbs the exact membership of the highest-ranked links, while the overall ordering structure remains strongly correlated. Taken together, these results indicate that observational noise compresses the effective length of the safely valid linear interval and weakens ranking consistency to some extent, but does not alter either the existence of the RLGI or the strict linearity of the resilience gain mechanism within the interval itself.

5.3.3. Cross-network validation on the Anaheim network

The above analyses establish the internal robustness of the proposed framework within the Nanjing case. We next investigate whether the within-interval linear validity of the RLGI transfers to a network with substantially different topology and scale. Fig. 8 presents the cross-network validation results on the Anaheim network for three representative weekday periods and three representative weekend periods. In each panel, the horizontal axis denotes the resilience gain predicted analytically by the RLGI, whereas the vertical axis denotes the recomputed resilience gain. The dashed 45° line indicates exact agreement between the predicted and recomputed resilience gains. The green points denote cases in which the imposed link-quality improvement remains entirely within the theoretically derived RLGI, whereas the blue points include interventions that partially exceed this interval.

Across all six periods, the within-RLGI cases align almost perfectly with the 45° line. For the $n = 50$ within-interval setting, the strict val-

**Fig. 8.** Cross-network validation of the proposed RLGI framework on the Anaheim network.

identity rate and the demand identity match rate both reach 1.0 in all tested periods, the mean reachability change remains zero, and the mean absolute estimation gap stays at the 10^{-17} level, confirming the exactness of the proposed closed-form formulation when the theoretical interval conditions are satisfied.

By contrast, once out-of-interval cases are included $n = 100$, visible deviations from the diagonal emerge, indicating that the resilience response is no longer governed by the strict linear relationship established within the RLGI. In this case, the resilience gain enters a more complex nonlinear regime. Taken together, these results support the transferability of the within-interval linear validity of the proposed framework to a network with substantially different topology and scale from the Nanjing case.

More broadly, the cross-network validation also suggests that differences in network structure may influence where the resilience-critical links emerge. The contrast between the Nanjing and Anaheim networks indicates that variations in topology and scale do not undermine the within-interval linear validity of the proposed RLGI framework. Rather, such differences are more likely to affect where critical links and nodes emerge and how concentrated the resulting resilience gains may be.

5.3.4. Computational efficiency and runtime analysis

Beyond robustness and transferability, practical applicability also depends on whether the analytical formulation can substantially reduce the computational burden relative to repeated equilibrium re-solving. Table 9 presents a direct runtime comparison between the proposed analytical framework and the UE-based numerical verification on the Anaheim network. The proposed framework requires only three computational stages, namely baseline UE computation, critical-link identification, and analytical RLGI evaluation, with a total runtime of 5.20 s. By contrast, the UE-based numerical verification requires an additional 101.68 s for repeated UE re-solving, resulting in a total runtime of 106.88 s. These results show that the proposed analytical framework itself is computationally lightweight, whereas the dominant runtime burden arises from repeated equilibrium re-solving introduced solely for external numerical validation. This computational contrast would be expected to become even more relevant in larger and operationally more complex urban networks, such as the Nanjing case, where repeated traffic reassignment is likely to be more burdensome than in a benchmark network like Anaheim. This distinction is practically important because reductions in incident duration are associated with lower delay and a reduced risk of secondary incidents in traffic incident management practice [72,73]. Against this background, the proposed analytical framework is better aligned with rapid operational decision-support requirements than repeated UE-based verification.

5.4. Identification of resilience-critical nodes and their implications

While link-level analysis reveals where resilience degradation is concentrated, network resilience is also strongly shaped by node characteristics arising from both static topological connectivity and the demand carried by connected links. To capture this dual role, we develop the CSDK-based method for resilience-critical node identification. Table 10 and Table 11 report the hourly calibrated parameters α and σ for May 29, which respectively balance the relative contributions of dynamic demand and static topology, and the trade-off between local node dominance and global network effects.

To evaluate the effectiveness of the proposed method, two comparative experiments are conducted. The first compares CSDK with two one-

dimensional strategies, namely the Connectivity-based and Demand-based methods, using unaffected demand (UD) and the largest connected component (LCC) as evaluation metrics. As shown in Fig. 9, the three strategies produce clearly different damage patterns during the morning peak (08:00–09:00), midday (12:00–13:00), and evening peak (18:00–19:00) on May 29.

At the 20 % node-removal threshold, the CSDK-based strategy causes substantially greater functional loss than the Connectivity-based strategy, while also inducing much stronger structural disruption than the Demand-based strategy. For example, unaffected demand under the CSDK-based attack drops to 23 %, 16 %, and 14 % across the three representative periods, compared with 61 %, 53 %, and 67 % under the Connectivity-based strategy. In terms of structural fragmentation, the CSDK-based method reduces the LCC to 36 %, 30 %, and 31 %, whereas the Demand-based strategy leaves 73 %, 76 %, and 70 % of the network still connected. These results indicate that the Connectivity-based method mainly damages static structure, whereas the Demand-based method mainly disrupts high-flow bottlenecks. By contrast, the proposed CSDK-based method simultaneously weakens both connectivity and demand support, thereby identifying nodes that are critical in a more balanced and system-wide sense.

The cumulative attack results further support this conclusion. Across all three periods, the area under the UD curve for the CSDK-based method (0.10–0.13) is approximately half that of the Connectivity-based method (0.22–0.26), indicating substantially stronger functional disruption. Meanwhile, the corresponding LCC areas remain comparable, suggesting that the CSDK-based method achieves much larger demand loss without sacrificing its ability to fragment the network structurally.

We further benchmark the CSDK-based method against six classical graph-theoretic indicators, including degree, betweenness, closeness, eigenvector centrality, PageRank, and the Collective Influence (CI) algorithm. As shown in Fig. 10, the CSDK-based strategy consistently produces the fastest decline in both UD and LCC across the three representative periods. When 20 % of nodes are removed, the CSDK-based attack disrupts approximately 80 % of total flow demand, whereas benchmark methods such as PageRank affect only about 44 %. Overall, these results demonstrate that the proposed method more effectively identifies nodes that are simultaneously critical to structural connectivity and traffic functionality.

5.5. Resilience recovery strategies

Building on the above node-level identification results, we next examine how these critical-node rankings can be translated into resilience recovery strategies under both daily disturbances and post-disaster scenarios.

(1) Recovery under daily disturbances

For short-horizon daily disturbances such as traffic collisions, localized pavement failures, and recurring congestion, rapid recovery is more important than large-scale re-optimization. Using May 29 as an example, the five most critical nodes are identified for the morning and evening peaks (Fig. 11a), and the corresponding recovery experiments are shown in Fig. 11b–c. The results indicate that restoring the top-ranked critical nodes identified by the CSDK-based framework accelerates resilience recovery more effectively than the benchmark approaches.

Table 9

Runtime comparison between the proposed analytical framework and the UE-based numerical verification on the Anaheim network.

Method	Baseline UE(s)	Critical-link identification(s)	Analytical RLGI (s)	UE re-solving (s)	Total runtime (s)
Proposed analytical framework	4.58	0.62	0.0013	0	5.20
UE-based numerical verification	4.58	0.62	0.0013	101.68	106.88

Table 10
Optimal α value on May29.

Time interval	6–7	7–8	8–9	9–10	10–11	11–12	12–13	13–14
α value	0.38	0.34	0.58	0.18	0.20	0.22	0.22	0.20
Time interval	14–15	15–16	16–17	17–18	18–19	19–20	20–21	21–22
α value	0.22	0.48	0.20	0.14	0.10	0.20	0.38	0.04

Table 11
Optimal sigma value on May 29.

Time interval	6–7	7–8	8–9	9–10	10–11	11–12	12–13	13–14
σ value	0.73	0.84	0.51	0.87	0.82	0.90	0.76	0.91
Time interval	14–15	15–16	16–17	17–18	18–19	19–20	20–21	21–22
σ value	0.89	0.53	0.82	0.92	0.96	0.82	0.73	0.68

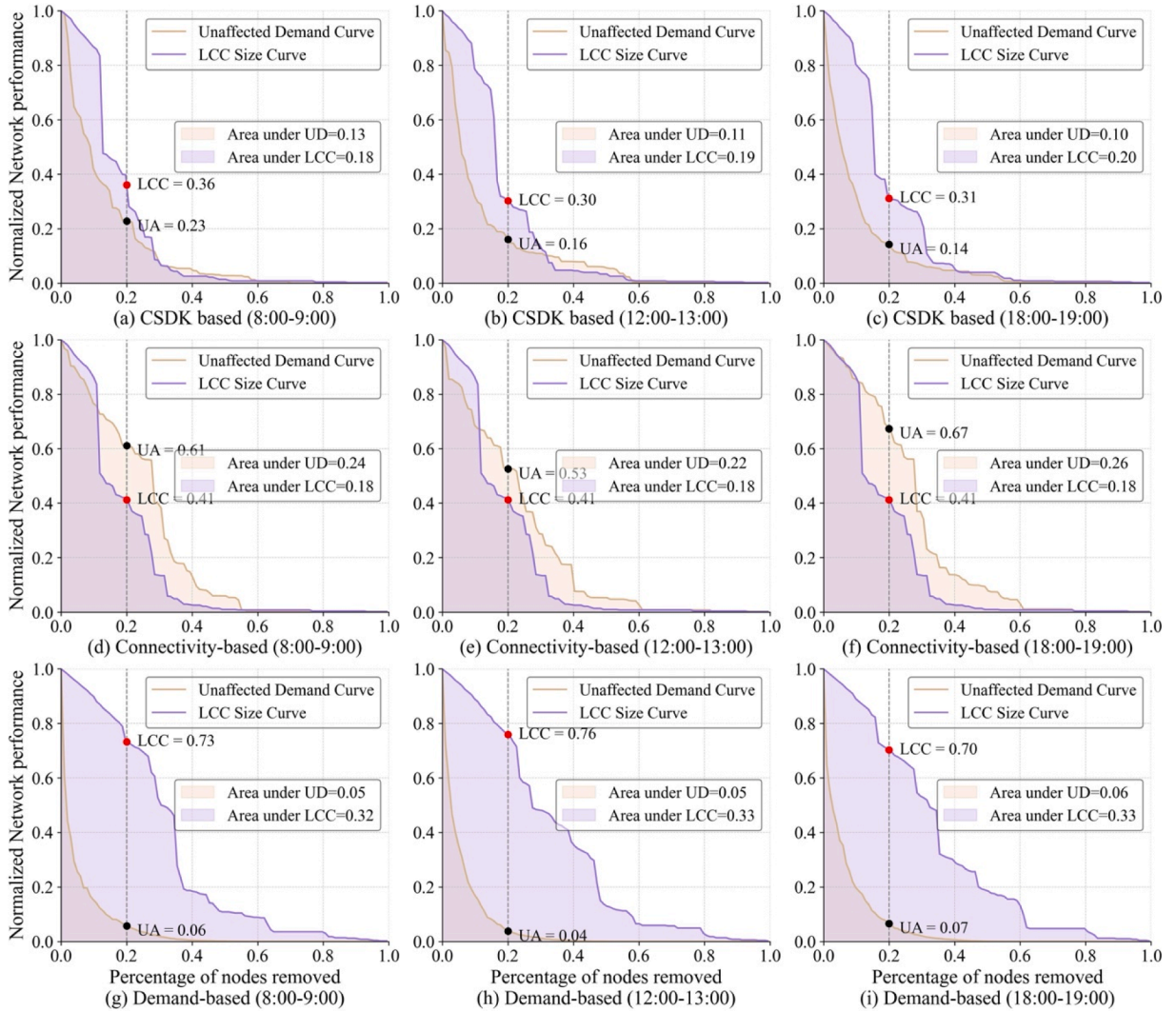


Fig. 9. Comparative performance of attack strategies on unaffected demand (UD) and largest connected component (LCC) for May 29. Colored polygons indicate the area under each curve; smaller areas correspond to greater damage. (The Fig. E of 31 May is presented in Appendix E).

Under the daily-disturbance setting, severe congestion on affected links is assumed to reduce link quality q_{ij}^t to zero, while OD demand and lwc_{ij}^t remain unchanged. These conditions satisfy the RLGI prerequisites, allowing resilience improvements to be evaluated directly using Eq. (10). The results show clear diminishing returns: the top-ranked node contributes the largest resilience improvement, while the marginal gain

decreases progressively for lower-ranked nodes. For the morning and evening peaks, the initial resilience values are 0.5955 and 0.6955, and the corresponding total resilience gains from restoring the selected critical nodes are 0.1681 and 0.1857, representing improvements of 28.23 % and 26.70 %, respectively. These findings provide both the optimal restoration order of critical nodes under minor disturbances and a direct estimate of their marginal recovery contributions.

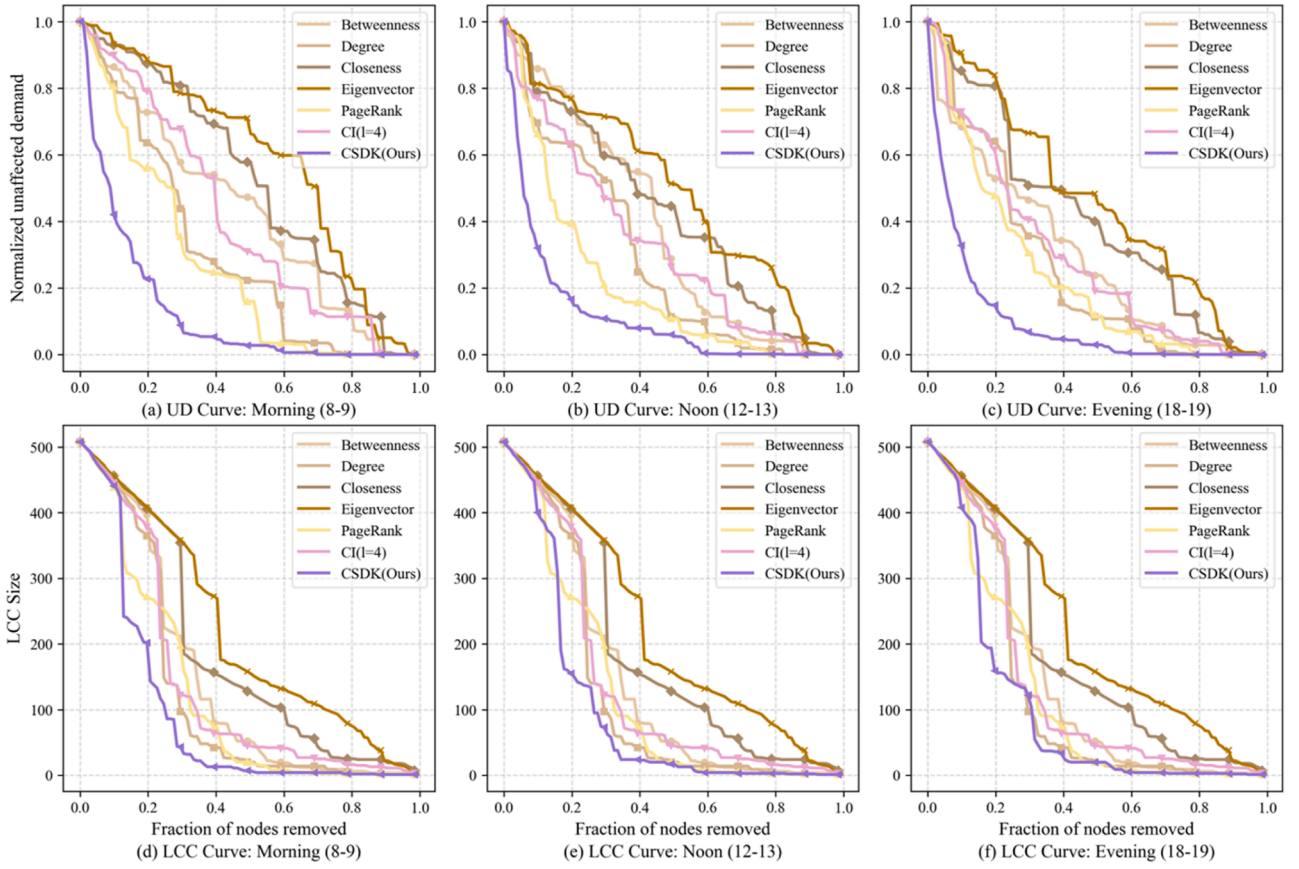


Fig. 10. Impact of attack strategies on network performance metrics UD and LCC for May 29 during 08:00–09:00 (morning peak), 12:00–13:00 (midday), and 18:00–19:00 (evening peak). (The Fig. F of 31 May is presented in Appendix F).

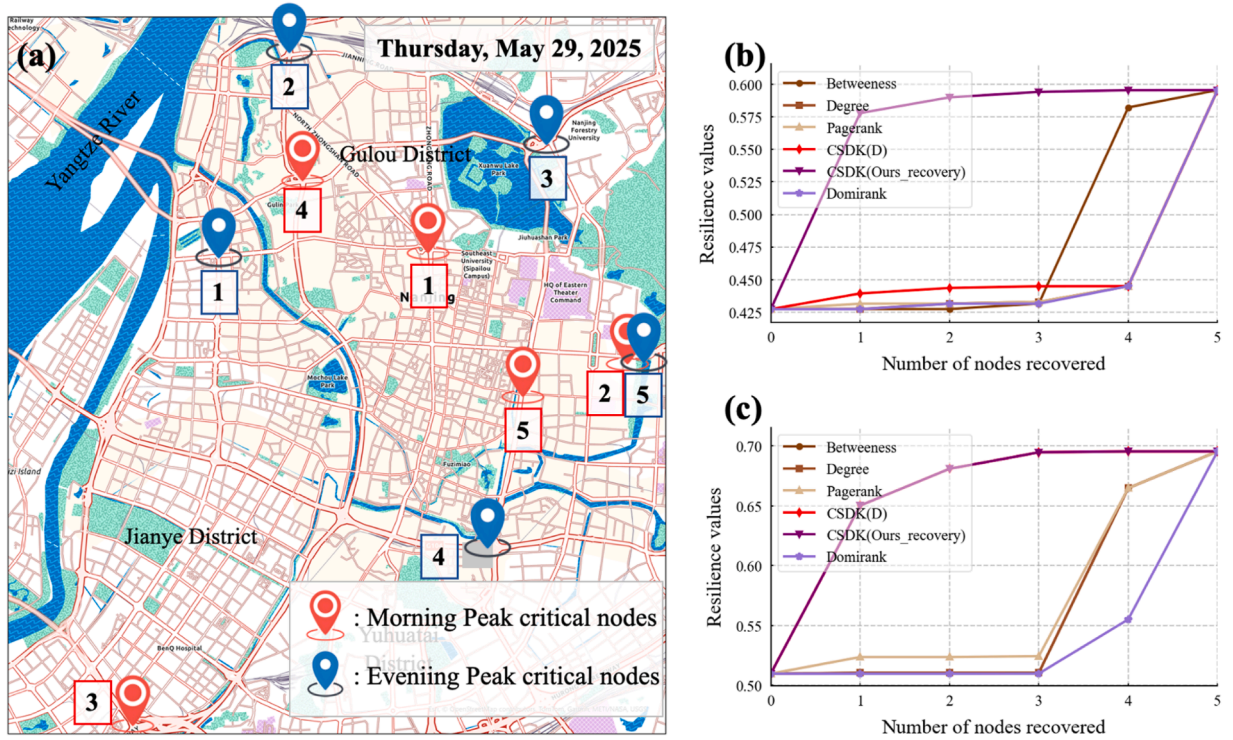


Fig. 11. Resilience recovery strategies for daily disruptions on May 29, 2025.

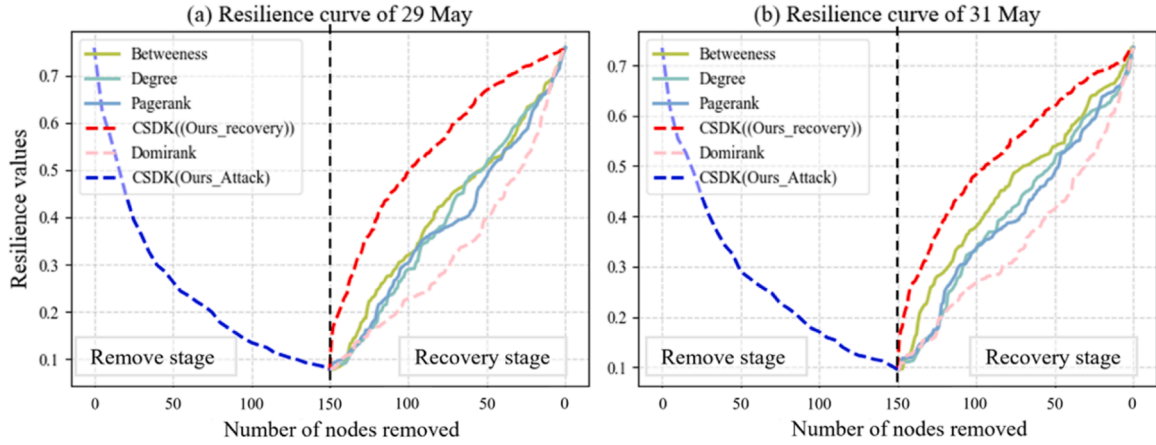


Fig. 12. Resilience evolution during node removal and recovery under four restoration strategies.

This result also shows that the analytical logic underlying resilience enhancement can be extended to short-horizon recovery evaluation, enabling rapid estimation of the resilience benefits associated with restoring the most critical nodes.

(2) Recovery under post-disaster disruptions

We further extend the analysis to long-term post-disaster recovery under large-scale network disruption. Specifically, the most critical 150 nodes are first removed using the CSDK-based attack strategy to generate a severe failure scenario, and the subsequent restoration process is then guided by different ranking strategies. Fig. 12 illustrates the resulting resilience evolution on May 29 (weekday) and May 31 (weekend), forming a typical resilience-triangle pattern.

During the restoration phase, the CSDK-based strategy consistently exhibits the steepest recovery trajectory, indicating faster resilience restoration than the PageRank-, degree-, betweenness-, and Domirank-based strategies. For example, under the CSDK-based strategy, network resilience exceeds 0.6 after approximately 250 nodes have been restored, whereas the benchmark methods require the restoration of nearly all removed nodes to reach similar levels. These results confirm that the proposed method more effectively prioritizes nodes that are jointly critical to structural connectivity and traffic functionality, thereby enabling a faster and more efficient post-disaster recovery process.

Taken together, these results show that the proposed CSDK-based framework not only identifies nodes that are jointly critical to structural connectivity and traffic functionality, but also translates this criticality into effective recovery priorities under both short-horizon daily disturbances and long-term post-disaster disruptions.

(3) Practical implementation under resource constraints

In practice, network recovery is often constrained by limited budgets, available repair crews, and restricted work windows. Although a formal budgeted optimization model is not explicitly solved in this study, the ranked recovery priorities and the associated resilience-top- k curves provide a direct and implementable decision rule. Under a given resource limit, decision-makers can restore the top- k critical nodes in phases, where k may represent the maximum number of nodes that can be repaired within a work shift or serve as a budget proxy under an equal-effort approximation. The observed diminishing returns further indicate the point beyond which additional restoration actions yield only limited marginal resilience gains, thereby supporting realistic target setting and transparent resource allocation. In practice, crew and logistical constraints may also favor restoring spatially clustered candidates to reduce mobilization time, or prioritizing distinct bottlenecks

to avoid redundant interventions when multiple critical nodes affect the same corridor.

6. Discussion

6.1. A novel daily resilience linear gain interval for link-quality improvements

Through rigorous theoretical derivation and empirical validation, this study is the first to establish a closed-form, piecewise-linear relationship that links the network-level resilience gain ΔR_t to the incremental quality improvement $\Delta q_{od,L}^{t,max}$ applied to a limiting link e_{ij} at hour t . As long as the improvement lies within the feasible interval $\Delta q_{od,L}^{t,max} \in [0, q_{safe} - q_{od,L}^{max}]$ the gain is strictly linear and obeys Eq. (10); once the threshold is exceeded, higher-order coupling effects dominate and the marginal benefit diminishes: a finding confirmed by the bifurcation in the resilience gain rate curves (Fig. 5 and Fig. 6). Moreover, existing pre-disaster resilience studies concentrate on providing additional route options or reconstructing critical infrastructure, yet they seldom quantify how much redundancy is added or how much resilience gain is achieved. Few researchers have introduced explicit equations for measuring network redundancy and translating it into resilience improvement. Specifically, our method not only establishes a precise linear gain relationship between link-quality increments and resilience improvements but also systematically answers three critical questions for traffic managers: how, when, and which to strategically intervene to maximize resilience. The specific critical questions for traffic managers are as follows:

- (1) **How to enhance resilience:** By deriving a closed-form relationship between resilience R and link quality q (Eq. (10)) and proving both analytically and empirically that the gain is strictly linear within the feasible increment $\Delta q_{od,L}^{t,max}$ (Eq. (10)), we show that upgrading a single, highest-link weighted coefficient yields a predictable and proportional improvement in network resilience. Prioritizing this link keeps transportation agencies reliably operate within the linear regime, ensuring predictable and proportional resilience enhancements. These quantified resilience benefits provide clear expectations for authorities regarding the returns on investment from transportation infrastructures upgrades, highlighting the high potential for effective use of limited resources.
- (2) **When to apply resilience upgrades:** Detailed hourly analyses in Fig. 7(a) demonstrate that weekday morning (08:00–09:00) and evening (18:00–19:00) peaks represent the optimal intervention windows with gain rates of 29.31 % and 21.43 %, respectively. By

contrast, weekend resilience gains, though consistent, are comparatively modest due to more evenly dispersed travel patterns. Such insights enable traffic managers to pinpoint exactly when road quality upgrades will yield the greatest resilience gains, thereby scheduling interventions at optimal times and markedly improving resource allocation efficiency.

- (3) **Which link to prioritize:** Multi-link comparisons (Fig. 7(b) - (c)) clearly indicate that upgrading the link with the highest link weighted coefficient consistently deliver substantially greater resilience gains rate compared to upgrading secondary alternatives. For example, in Fig. 7(b), replacing an upgrade to the most critical link (CI-1) with one to the second-ranked link (CI-2) during the morning peak reduces the resilience gain rate from about 29.31 % to just 9.08 %. This sharp drop underscores the strategic value of directing limited resources to the single most critical link, thereby helping traffic authorities accurately determine which road segment to upgrade and maximizing both efficiency and resilience gains.

Taken together, our framework provides not only theoretical insights but also highly actionable strategic guidance for urban traffic resilience enhancement. By clearly delineating the optimal timing, locations, methods, and magnitude of resilience interventions, transportation planners and authorities are empowered with a robust decision-support tool. Practically, this translates to reduced congestion-related delays, improved response capacity during incidents, and significantly strengthened overall operational stability and reliability of the urban transportation network. Moreover, these insights can inform longer-term infrastructure planning decisions, aiding in designing resilient road networks capable of effectively absorbing and recovering from both daily traffic fluctuations and unforeseen disruptions.

6.2. Limitations and future work

Despite the encouraging findings, several limitations should be acknowledged, which also motivate important directions for future work. First, this study focuses on the urban road transportation subsystem (URTS-road) and does not explicitly model metro, bus, or other modes, nor mode shifts and transfer behaviors. In addition, although cross-network validation has been conducted on the Anaheim network, broader validation across multiple real-world cities with diverse urban structures and traffic conditions remains an important direction for future research. Future research will extend the framework to multi-modal settings in large scale cities to capture intermodal interactions and demand reallocation across modes under disturbances and recovery actions. Second, the road topology is constructed from OpenStreetMap (OSM) and then simplified to major road classes. While this supports transparency and reproducibility, OSM may contain incomplete or inaccurate attributes for certain links, and topology simplification may omit local streets, potentially affecting fine-scale link representation. Future work will incorporate authoritative GIS layers and richer operational attributes, and will examine the sensitivity of conclusions to alternative mapping and network-construction choices. Finally, beyond the RLGI regime, resilience gains exhibit nonlinear behavior associated with structural transitions and network-wide coupling; future work will further investigate these mechanisms and extend the framework to hazard-driven scenarios (e.g., heavy rainfall) where disruptions are more severe and spatially correlated.

7. Conclusions

With rapid urban expansion and intensifying climate volatility, urban road transportation systems (URTS) are increasingly exposed to both recurrent daily disturbances and large-scale disruptions. This study develops an integrated resilience analysis framework that unifies resilience assessment, link- and node-level influence analysis, and recovery

evaluation within a single process, thereby supporting more effective resilience management of urban road systems under diverse disturbance conditions.

We establish a time-varying resilience assessment approach that captures fluctuations in urban road networks during daily disturbances. Unlike conventional static post-disaster measures, this approach reveals how resilience evolves continuously in response to daily yet persistent disruptions, offering a diagnostic foundation for systematic analysis. Building on this basis, empirical evidence from large-scale observations in Nanjing uncovers a consistent law: incremental link-quality improvements yield proportional gains in network resilience within a Resilience Linear Gain Interval, while exceeding this range leads to diminishing returns. This linear gain mechanism further demonstrates strong robustness, transferability, and practical applicability. It remains stable across the full 16-day sample, across weekday and weekend conditions, across different demand heterogeneity regimes, and under observational noise. Cross-network validation on the Anaheim network and runtime comparisons further support its transferability and efficiency for rapid operational use.

Extending beyond link-level interventions, the identification of critical nodes through the CSDK-based method integrates link demands and topological connectivity, outperforming traditional benchmarks by pinpointing nodes whose disruption most severely undermines network resilience. Finally, recovery analysis highlights predictable pathways: under daily disturbances, rapid restoration of top-ranked critical nodes yields the greatest immediate resilience gains, whereas under large-scale disruption scenarios, recovery requires prioritized restoration of structurally and functionally critical nodes.

Together, these results move resilience research beyond descriptive evaluation toward a quantitative and actionable framework for daily assessment, targeted intervention, and recovery in urban road networks under both routine and large-scale disruptions. This provides practical support for more effective daily operation and post-disruption recovery planning.

Data availability

16 days of Nanjing's road transportation network data used in this study, are available at <https://doi.org/10.5281/zenodo.17218472>.

Code availability

Source codes for the algorithms proposed in this study are available at <https://github.com/Nor-Eich/Daily-resilience-of-urban-road-transportation-systems-assessment-enhancement-and-recovery>.

CRediT authorship contribution statement

Qi Ai: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Chengcheng Xu:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Hao Tong:** Writing – review & editing, Data curation, Conceptualization. **Chang Peng:** Writing – review & editing, Investigation, Formal analysis. **Haosong Wen:** Writing – review & editing, Methodology. **Michael Levin:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. List of notations

Table 1

List of notations.

Notations	Explanations
t	Time
$G(t)$	An undirected topological graph of the urban road transportation network
V	The set of intersection nodes in urban road transportation network
E	The set of links in urban road transportation network
A	The adjacent matrix of urban road transportation network
H^t	The traffic network weight matrix during hour t
W	The criticality-based link weight matrix
F, f_{od}^t	The demand matrix, the flow demand between node o to node d during hour t
Ψ_{od}, ψ	All paths connecting o to d , an undirected path
$D, D_{\mathcal{A}}, D_{\mathcal{B}}, D_{\mathcal{C}}$	The set of all origin-destination pairs with non-zero flow demand, the set of limiting-link pairs, the set of suboptimal-path bottleneck pairs, the set of unaffected pairs
De, De_{ρ}^t	De represents the sum of flow for all OD pairs in the network, De_{ρ}^t denotes the sum of flow for OD pairs which can remain served during the percolation process.
N, N_{GC}	N_{LGC} represents the number of nodes in the giant component, and N is the number of all nodes in the URTS network
$B(\Psi, t)$	The bottleneck link of path ψ
Q, q_{ij}	The link quality matrix, the link quality of link between node i and node j
$q_{od,t}^{max}, q_{B_{\psi}}^{min}$	The quality of the limiting link $L_{od} \in E$ for any pair (o, d) with non-zero f_{od} during hour t , the quality of the bottleneck link.
q_{ψ}	The second smallest quality of the path ψ after removing the limiting link L_{od} .
q_{safe}	The maximum quality that the link can achieve within the Safe Linear Interval.
$v_{ij}(t)$	the average speed on link e_{ij} during hour t
ϱ_{od}^t	Set of optimal bottleneck paths for OD pair (o, d) at interval t
ρ	The percolation threshold
UD_{ρ}	The proportion of total origin-destination (O-D) demand that remains unaffected when the percolation threshold is set to ρ
lwc_{ij}^t	The weighted coefficient of link (i, j) during hour t
$R, \Delta R$	The resilience of urban road transportation system, the change in resilience
σ	The intensity of competition
Γ	The DomRank score
$\alpha, \beta, \theta, \lambda$	The optimal parameter
a, b	The parameters that balance the influence of flow demand and network connectivity
a_t^*, b_t^*	The best parameters that balance dynamic flow demand during hour t and static network topology
UD_{AUC}	The area under the normalized size of the unaffected demand curve UD_{ρ} versus ρ
LCC_{AUC}	The area under the normalized size of the largest connected component LCC_{ρ} versus ρ

Appendix B. Proof of the Resilience Linear Gain Interval (RLGI)

Resilience Linear Gain Interval (RLGI)

This appendix provides the theoretical basis of the Resilience Linear Gain Interval (RLGI) under the revised bottleneck-path formulation. The key idea is that, as long as a link-quality improvement does not alter the optimal bottleneck-path structure, the corresponding bottleneck allocation logic remains unchanged, and the resulting resilience gain is strictly linear in the link-quality increment.

The resilience over the time window $[t_1, t_2]$ is defined in the main text as:

$$R_{[t_1, t_2]} = \frac{1}{t_2 - t_1} \sum_{t=t_1}^{t_2-1} \sum_{e_{ij} \in E} lwc_{ij}^t \cdot q_{ij}^t. \quad (B.1)$$

Now consider an intervention that improves link quality by Δq_{ij}^t . If the intervention remains within the RLGI, then for every affected interval t and active OD pair (o, d) , the following objects remain unchanged: (1) the optimal bottleneck value $q_{od}^{(t,*)}$, (2) the optimal bottleneck-path set ϱ_{od}^t , (3) the bottleneck-link set $B(\psi, t)$ for every $\psi \in \varrho_{od}^t$.

Under this condition, the bottleneck allocation rule remains unchanged, and therefore the corresponding link weighted coefficients lwc_{ij}^t also remain unchanged. Hence, after the intervention, the resilience can be written as:

$$R'_{[t_1, t_2]} = \frac{1}{t_2 - t_1} \sum_{t=t_1}^{t_2-1} \sum_{e_{ij} \in E} lwc_{ij}^t (q_{ij}^t + \Delta q_{ij}^t). \quad (B.2)$$

Subtracting Eq. (B.1) from Eq. (B.2) yields:

$$\Delta R_{[t_1, t_2]} = R'_{[t_1, t_2]} - R_{[t_1, t_2]} = \frac{1}{t_2 - t_1} \sum_{t=t_1}^{t_2-1} \sum_{e_{ij} \in E} lwc_{ij}^t \cdot \Delta q_{ij}^t. \quad (B.3)$$

Equation (B.3) shows that, within the RLGI, the resilience gain is a linear function of the link-quality improvement. In the special case where only one link e_{ij} is improved at one interval t , Eq. (B.3) reduces to:

$$\Delta R_{[t_1, t_2]} = \frac{lwc_{ij}^t}{t_2 - t_1} \cdot \Delta q_{ij}^t, \quad (\text{B.4})$$

which implies that the slope of the resilience response is:

$$\frac{\partial R_{[t_1, t_2]}}{\partial q_{ij}^t} = \frac{lwc_{ij}^t}{t_2 - t_1}, \quad (\text{B.5})$$

provided that the intervention remains within the RLGI.

Therefore, the RLGI provides a valid regime in which resilience gains can be computed directly from link-quality improvements without repeated traffic reassignment. Once the intervention exceeds this interval, the optimal bottleneck-path structure may change, and the corresponding lwc_{ij}^t must be recomputed.

To operationalize this property, the following algorithm determines the largest admissible quality value of a candidate link such that the optimal bottleneck-path structure remains unchanged.

Algorithm 1

Resilience linear gain interval boundaries.

```

1:      Input: candidate link  $e$ , link-quality matrix  $Q^t$ , active OD set  $W^t$ , feasible path set  $\Psi_{od}^t$ 
2:      Output: safe upper bound  $q_{safe}(e, t)$ 
3:      Initialize  $q_{safe}(e, t) \leftarrow 1$ 
4:      For each  $(o, d) \in W^t$  do:
5:          Compute the current optimal bottleneck value:
6:           $q_{od}^{(t,*)} \leftarrow \max_{\psi \in \Psi_{od}^t} q_{\psi}^t$ 
7:          Compute the current optimal bottleneck-path set:
8:           $\mathcal{Q}_{od}^t \leftarrow \arg \max_{\psi \in \Psi_{od}^t} q_{\psi}^t$ 
9:          If  $e \in B(\psi, t)$  for some  $\psi \in \mathcal{Q}_{od}^t$  then:
10:             Initialize the admissible bound for OD pair  $(o, d)$  as  $q_{od}^{safe}(e, t) \leftarrow 1$ 
11:             Increase the quality of link  $e$  gradually from  $q_e^t$ :
12:             For each candidate improved value  $\tilde{q}_e^t$  do:
13:                 Recompute  $q_{\psi}^t$  for all  $\psi \in \Psi_{od}^t$  affected by link  $e$ 
14:                 Recompute the updated optimal bottleneck value  $\tilde{q}_{od}^{(t,*)}$ 
15:                 Recompute the updated optimal bottleneck-path set  $\tilde{\mathcal{Q}}_{od}^t$ 
16:                 Recompute the updated bottleneck-link sets  $\tilde{B}(\psi, t)$  for  $\psi \in \tilde{\mathcal{Q}}_{od}^t$ 
17:                 If  $(\tilde{q}_{od}^{(t,*)} \neq q_{od}^{(t,*)}) \vee (\tilde{\mathcal{Q}}_{od}^t \neq \mathcal{Q}_{od}^t) \vee (\exists \psi, \tilde{B}(\psi, t) \neq B(\psi, t))$  then:
18:                     break
19:                 Else:
20:                     Update  $q_{od}^{safe}(e, t) \leftarrow \tilde{q}_e^t$ 
21:             End if
22:         End for
23:         Update  $q_{safe}(e, t) \leftarrow \min\{q_{safe}(e, t), q_{od}^{safe}(e, t)\}$ 
24:     End if
25: End for
26: Return  $q_{safe}(e, t)$ 

```

Note: Ψ_{od}^t denotes the set of all feasible directed paths for OD pair (o, d) at interval t . $q_{\psi}^t = \min_{e_{ij} \in \psi} q_{ij}^t$ is the bottleneck quality of path ψ . $\mathcal{Q}_{od}^t$ denotes the optimal bottleneck-path set, and $B(\psi, t)$ denotes the set of bottleneck links on path ψ .

For a candidate link improvement to remain within the RLGI, the optimal bottleneck value $q_{od}^{(t,*)}$, the optimal path set $\mathcal{Q}_{od}^t$, and the associated bottleneck-link sets $B(\psi, t)$ must all remain unchanged. The returned $q_{safe}(e, t)$ is the largest admissible quality value such that any improvement satisfying:

$$q_e^t + \Delta q_e^t \leq q_{safe}(e, t) \quad (\text{B.6})$$

remains within the RLGI.

Once the interval is identified, resilience gains can be computed directly using Eq. (10) in the main text without repeated UE re-solving, thereby enabling rapid identification of priority links for resilience enhancement.

Appendix C. Algorithm 2: Linear Enhancement under Cost Constraints

Within the Resilience Linear Gain Interval (RLGI), we have shown that network resilience R responds linearly to improvements in link quality q , with slope determined by the link's weighted coefficient lwc . This property allows us to bypass computationally intensive traffic assignment or simulation models and instead compute resilience gains directly. To translate this into practice, we design a cost-constrained enhancement algorithm that prioritizes links based on their marginal contribution to resilience.

Algorithm 2

Linear enhancement.

```

1:      Input: limiting link set  $L$ , coefficient  $lwc_L$  and quality  $q_L^{max}$ , path set  $\Psi_{od}$ , demand set  $D$ , target cost  $TC$ 
2:      Output: resilience improvement  $\Delta R$ , the quality improvement  $\Delta q$  and resilience gain  $\frac{\Delta R}{\Delta q}$ 
3:      Sort  $L$  in descending order of  $lwc_L$ 
4:      Initialize  $\Delta R \leftarrow 0$ 
5:      For each  $L_{od}$  in  $L$  do
6:          If  $TC = 0$ : break
7:           $\delta q \leftarrow \min(\min(q'_{od}, q''_{od}, 1) - q_{od,L}^{max}, TC)$ 
8:           $\Delta q \leftarrow \delta q$ 
9:           $\Delta R \leftarrow \Delta R + \frac{lwc_L}{t_2 - t_1} \Delta q$ 
10:          $TC = TC - \delta q$ 
11:          $\frac{\Delta R}{\Delta q} \leftarrow \frac{\Delta R}{\Delta q}$ 
12:     End for
13:     Return  $\Delta q, \Delta R, \frac{\Delta R}{\Delta q}$ 

```

Appendix D. Detailed formulation and calibration of the CSDK-based critical node identification framework**D.1 Domirank-based competitive dynamics**

To quantify node-level dominance in a traffic-weighted network, the proposed CSDK framework builds on a Domirank-based competitive dynamics model. Let $\Gamma(t) \in \mathbb{R}^{N \times 1}$ denote the node dominance score vector at time t . Its temporal evolution is described as follows:

$$\frac{d\Gamma(t)}{dt} = \alpha H^t (\theta \mathbf{1} * N \times \mathbf{1} - \Gamma(t)) - \beta \Gamma(t), \quad (D.1)$$

where $H^t \in \mathbb{R}^{N \times N}$ is the time-varying traffic weight matrix during hour t , $\mathbf{1} * N \times \mathbf{1}$ is an all-one vector and $\alpha, \beta, \theta \in \mathbb{R}^+$. Here, α controls the strength of competitive interaction, β governs dissipation, and θ acts as a threshold parameter that scales the equilibrium dominance level.

Following Engsig et al. [74], we define

$$\sigma = \frac{\alpha}{\beta}. \quad (D.2)$$

Since the parameter θ only rescales Γ without changing the relative node ranking, and β can be absorbed into the ratio σ , we set $\beta = 1$ without loss of generality. Under this setting, σ becomes the key control parameter that balances local node-level dominance and global network-structure effects. Directly solving the stationary state of Eq. (D.1) through matrix inversion can be computationally expensive for large sparse urban road networks. To improve numerical efficiency, we adopt a recursive update form of the CSDK centrality:

$$\Gamma(t + \Delta t) = \Gamma(t) + (\sigma H^t (\theta \mathbf{1} - \Gamma(t)) - \Gamma(t)) \Delta t. \quad (D.3)$$

Starting from an initial score vector, Eq. (D.3) is iteratively updated until convergence to the stationary solution Γ . The converged dominance score is then used to rank resilience-critical nodes. In this way, node importance is derived from a competitive process defined over the traffic-weighted network rather than from static topology alone.

D.2 Calibration of the parameters**D.2.1 Calibration of the weighting parameters (a, b)**

To calibrate the weighting parameters (a, b) in Eq. (11), each candidate traffic weight matrix is evaluated using two complementary percolation-based criteria: the unaffected-demand area under the curve (*UD.AUC*) and the largest-connected-component area under the curve (*LCC.AUC*). To determine the optimal trade-off between dynamic flow demand and static network topology, we sample $a \in [0, 1]$ at regular intervals and set $b = 1 - a$. For each candidate pair (a, b), the CSDK method is applied to compute node dominance scores, after which percolation simulations are performed to evaluate (*UD.AUC*) and (*LCC.AUC*). The details are as following:

In our framework, an OD demand is considered unaffected if at least one feasible path remains connected under a given percolation threshold ρ . For each OD pair, feasible paths are identified in the intact network using Dijkstra's algorithm. During the percolation process, an OD demand f_{od}^t is counted as unaffected if at least one feasible path remains connected after applying the threshold ρ ; otherwise, it is classified as affected.

Accordingly, the normalized unaffected demand is defined as

$$UD_\rho^t = \frac{De_\rho^t}{De} \quad (D.4)$$

De is the total demand across all OD pairs and De_ρ^t is the total demand of OD pairs that remain connected under threshold ρ . The relative size of the giant component is defined as

$$LCC = \frac{N_{LCC}}{N} \quad (D.5)$$

N_{LCC} is the number of nodes in the largest connected component and N is the total number of nodes in the network. By varying the threshold ρ , the curves UD_ρ and LCC_ρ are obtained, and their corresponding areas under the curve are denoted as UD_AUC and LCC_AUC , respectively.

Because maximizing UD_AUC and maximizing LCC_AUC are conflicting objectives, we apply non-dominated sorting to the candidate set (a, UD_AUC, LCC_AUC) . Any candidate dominated on both criteria by another candidate is discarded, leaving only the Pareto-optimal set. From this Pareto-optimal set, the final parameter pair (a_t^*, b_t^*) is selected using the knee-point rule, defined as the point with the maximum perpendicular distance to the line connecting the two extreme Pareto-optimal points. This selected pair is then used to construct the final traffic weight matrix H^t for hour t .

D.2.2 Calibration of σ

The parameter σ controls the balance between local node dominance and global network effects in the Domirank dynamics. A lower σ emphasizes local competitive dominance, whereas a higher σ strengthens global network-structure effects.

For each hourly network, σ is calibrated by searching its feasible interval and selecting the value that yields the best evaluation performance. In implementation, a candidate set of σ values is generated within the admissible interval, and the CSDK procedure is repeated for each candidate. The final value σ_t^* is chosen according to the evaluation criterion defined in the main text. In this way, both the weighting parameters (a, b) and the dominance parameter σ are determined in a data-driven manner and allowed to vary across time periods.

Appendix E. Comparative performance of attack strategies on unaffected demand (UD) and largest connected component (LCC) for May 31

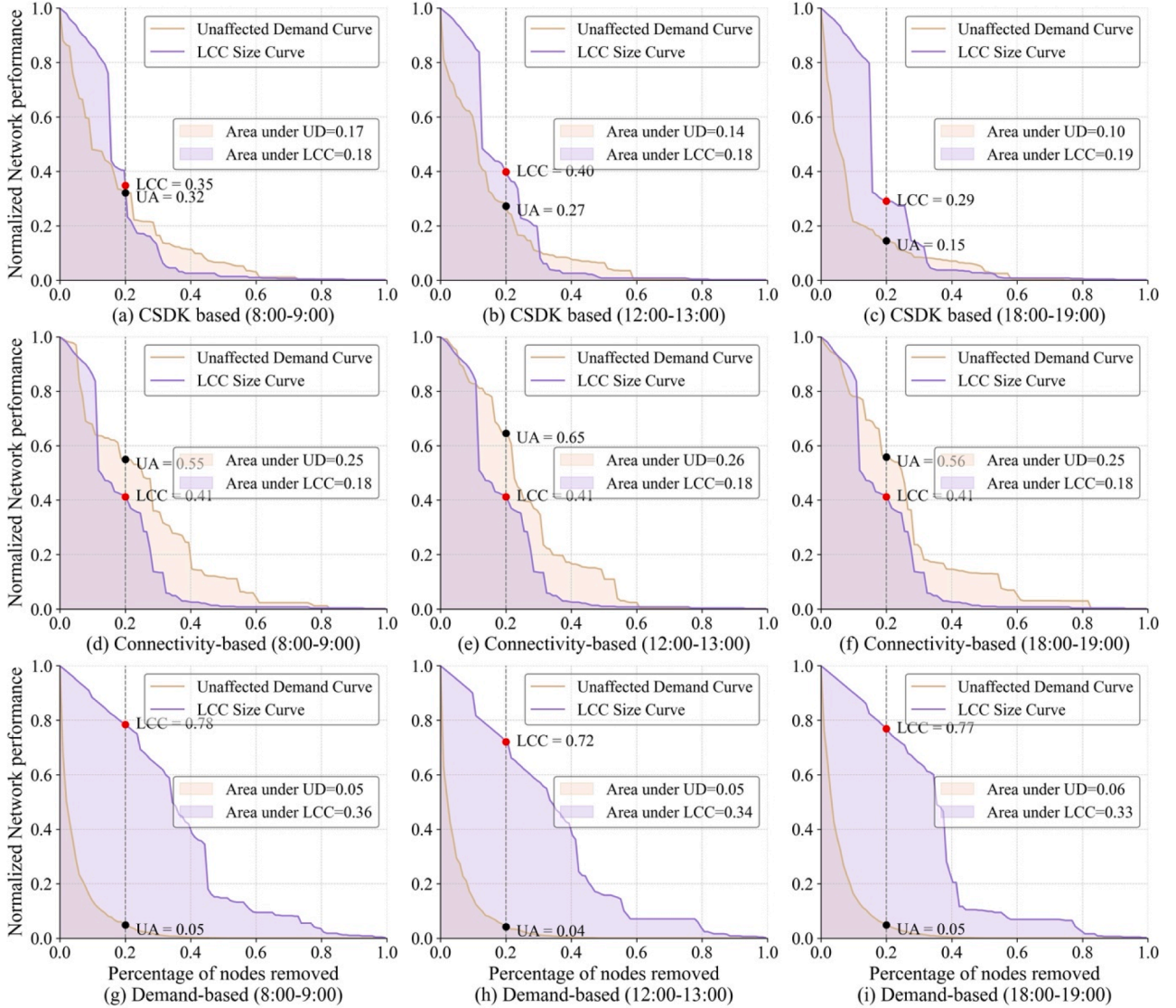


Fig. E. Comparative performance of attack strategies on unaffected demand (UD) and largest connected component (LCC) for May 31. Fig. D.1 (a)-(c) show results for CSDK-based method during 08:00–09:00 (morning peak), 12:00–13:00 (midday), and 18:00–19:00 (evening peak). Fig. D.1 (d)-(f) show results for the Domirank-based method during the same periods. Fig. D.1 (g)-(i) show results for the Demand-based method during the same periods. Colored polygons indicate the area under each curve (The smaller the area is, the greater the damage is).

Appendix F. Impact of attack strategies on network performance metrics UD and LCC for May 31

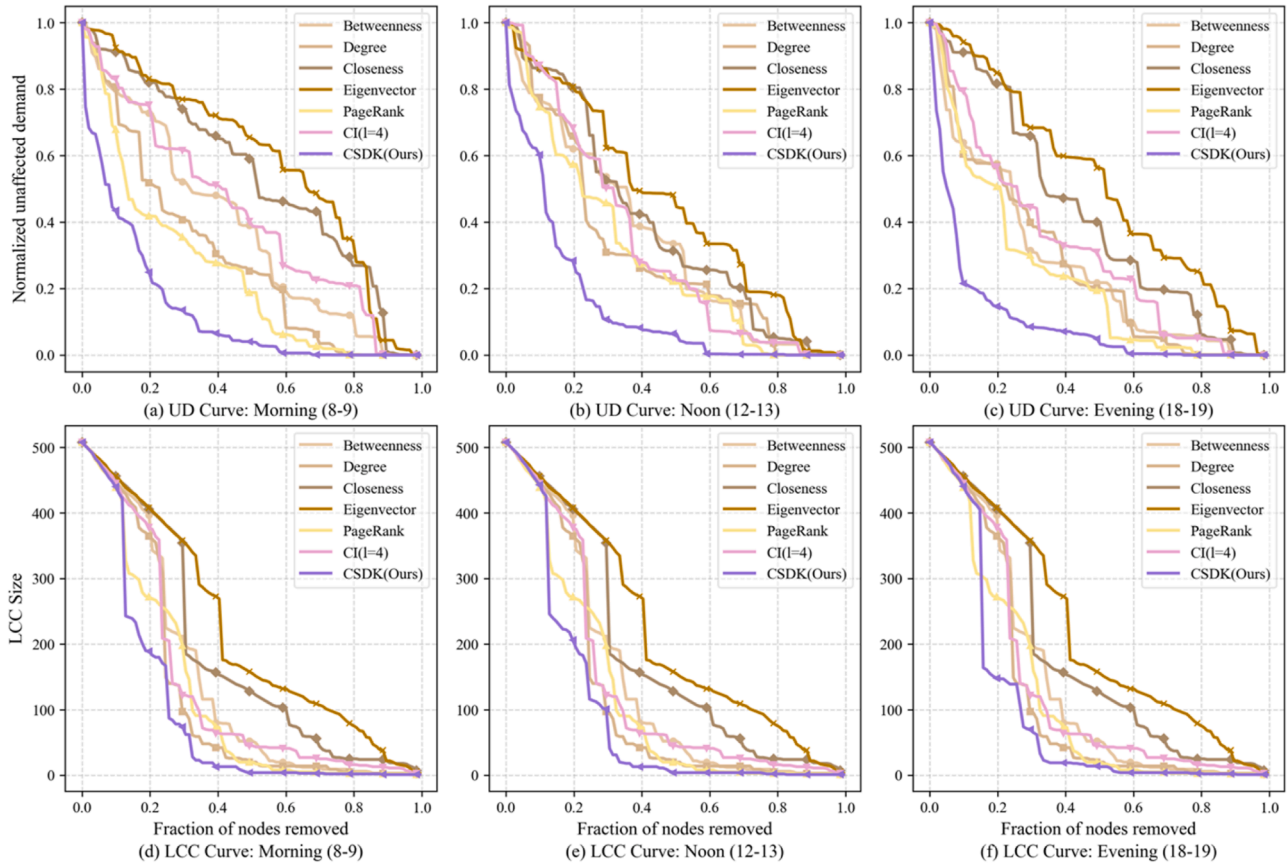


Fig. F. Impact of attack strategies on network performance metrics UD and LCC for May 31 during 08:00–09:00 (morning peak), 12:00–13:00 (midday), and 18:00–19:00 (evening peak).

Data availability

I have shared the link to my data and code in the manuscript.

References

- [1] Zhu J, Xu X, Wang Z. Economic evaluation of redundancy design for transportation networks under disruptions: framework and case study. *Transp Policy (Oxf)* 2023; 142:70–83. <https://doi.org/10.1016/j.tranpol.2023.08.004>.
- [2] Liu X, Chen J, Du B, Yan X, Peng Q, Shen J. Resilience assessment of urban rail transit stations considering disturbance and time-varying passenger flow. *Comput Aided Civ Eng* 2025;40:1627–51. <https://doi.org/10.1111/mice.13400>.
- [3] Guo J, Du Q, He Z. A method to improve the resilience of multimodal transport network: location selection strategy of emergency rescue facilities. *Comput Ind Eng* 2021;161:107678. <https://doi.org/10.1016/j.cie.2021.107678>.
- [4] Bi W, Schooling J, MacAskill K. Assessing flood resilience of urban rail transit systems: complex network modelling and stress testing in a case study of london. *Transp Res D: Transp Environ* 2024;134:104263. <https://doi.org/10.1016/j.trd.2024.104263>.
- [5] Ma Z, Yang X, Chen A, Zhu T, Wu J. Assessing the resilience of multi-modal transportation networks with the integration of urban air mobility. *Transp Res A: Policy Pract* 2025;195:104465. <https://doi.org/10.1016/j.tra.2025.104465>.
- [6] Zhang L, Chen T, Liu Z, Yu B, Wang Y. Analysis of multi-modal public transportation system performance under metro disruptions: a dynamic resilience assessment framework. *Transp Res A: Policy Pract* 2024;183:104077. <https://doi.org/10.1016/j.tra.2024.104077>.
- [7] Chen J, Liu J, Du B, Peng Q, Yin Y. Resilience assessment of an urban rail transit network under short-term operational disturbances. *IEEE Trans Intell Transp Syst* 2022;23:24841–53. <https://doi.org/10.1109/TITS.2022.3195937>.
- [8] Liu Z, Yu B, Zhang L, Sun Y. Resilience enhancement of multi-modal public transportation system via electric bus network redesign. *Transp Res E: Logist Transp Rev* 2025;193:103810. <https://doi.org/10.1016/j.tre.2024.103810>.
- [9] Gokalp C, Patil PN, Boyles SD. Post-disaster recovery sequencing strategy for road networks. *Transp Res B: Methodol* 2021;153:228–45. <https://doi.org/10.1016/j.trb.2021.09.007>.
- [10] Aghababaei M, Costello SB, Ranjithkar P. Transportation impact assessment following a potential alpine fault earthquake in new zealand. *Transp Res D Transp Environ* 2020;87:102511. <https://doi.org/10.1016/j.trd.2020.102511>.
- [11] Zhang M, Yang X, Zhang J, Li G. Post-earthquake resilience optimization of a rural “road-bridge” transportation network system. *Reliab Eng Syst Saf* 2022;225: 108570. <https://doi.org/10.1016/j.res.2022.108570>.
- [12] Mattsson L-G, Jenelius E. Vulnerability and resilience of transport systems – a discussion of recent research. *Transp Res Policy Pr* 2015;81:16–34. <https://doi.org/10.1016/j.tra.2015.06.002>.
- [13] Carrion C, Levinson D. Value of travel time reliability: a review of current evidence. *Transp Res Policy Pr* 2012;46:720–41. <https://doi.org/10.1016/j.tra.2012.01.003>.
- [14] Wang J, Wang C, Lv J, Zhang Z, Li C. Modeling travel time reliability of road network considering connected vehicle guidance characteristics indexes. *J Adv Transp* 2017;2017:1–9. <https://doi.org/10.1155/2017/2415312>.
- [15] Dong X, Cui H, Su Y, Zhu M, Yao S. Identifying critical road segments and optimizing resilience strategies based on multi-state congested characteristics. *Reliab Eng Syst Saf* 2025;258:110912. <https://doi.org/10.1016/j.res.2025.110912>.
- [16] Saadat Y, Ayyub BM, Zhang Y, Zhang D, Huang H. Resilience-based strategies for topology enhancement and recovery of metrorail transit networks. *ASCE-ASME J Risk Uncertain Eng Syst A: Civ Eng* 2020;6. <https://doi.org/10.1061/ajrua6.0001057>.

- [17] Tran VN, Ivanov VY, Huang W, Murphy K, Daneshvar F, Bednar JH, et al. Connectivity in urban landscapes can cause unintended flood impacts from stormwater systems. *Nat Cities* 2024;1:654–64. <https://doi.org/10.1038/s44284-024-00116-7>.
- [18] Padgett JE, Rincón R, Panakal P. Future cities demand smart and equitable infrastructure resilience modeling perspectives. *NPJ Nat Hazards* 2024;1:28. <https://doi.org/10.1038/s44304-024-00028-5>.
- [19] Guo L, Jiang C, Hou W, Ng AKY, Shi Q. International multimodal transport connectivity assessment of multimodal transport from mainland China to Europe. *Transp Res E: Logist Transp Rev* 2024;186:103564. <https://doi.org/10.1016/j.tre.2024.103564>.
- [20] Ai Q., Xu C., Tong H., Wen H., Peng C., Levin M. An integrated resilience dynamic assessment framework for urban road transportation systems under daily disruptions 2026. <https://doi.org/10.1016/j.jclepro.2026.147987>.
- [21] Bruneau M, Chang SE, Eguchi RT, Lee GC, O'Rourke TD, Reinhorn AM, et al. A framework to quantitatively assess and enhance the seismic resilience of communities. *Earthq Spectra* 2003;19:733–52. <https://doi.org/10.1193/1.1623497>.
- [22] Akbarzadeh M, Memarmontazerin S, Derrible S, Salehi Reihani SF. The role of travel demand and network centrality on the connectivity and resilience of an urban street system. *Transportation (Amst)* 2019;46:1127–41. <https://doi.org/10.1007/s11116-017-9814-y>.
- [23] Tang J, Lin H, Fan X, Yu X, Lu Q. A topology-based evaluation of resilience on urban road networks against epidemic spread: implications for COVID-19 responses. *Front Public Health* 2022;10:1023176. <https://doi.org/10.3389/fpubh.2022.1023176>.
- [24] Twumasi-Boakye R, Sobanjo JO. Resilience of regional transportation networks subjected to hazard-induced bridge damages. *J Transp Eng A: Syst* 2018;144. <https://doi.org/10.1061/jteps.0000186>.
- [25] Zhou Y, Wang J, Yang H. Resilience of transportation systems: concepts and comprehensive review. *IEEE Trans. Intell Transp Syst* 2019;20:4262–76. <https://doi.org/10.1109/TITS.2018.2883766>.
- [26] Yang Y, Huang H, Li G, Han B, Yuan Z, Ma H. A systematic review of resilience assessment and enhancement of urban integrated transportation networks. *J Transp Geogr* 2025;129:104420. <https://doi.org/10.1016/j.jtrangeo.2025.104420>.
- [27] Barker K, Ramirez-Marquez JE, Rocco CM. Resilience-based network component importance measures. *Reliab Eng Syst Saf* 2013;117:89–97. <https://doi.org/10.1016/j.res.2013.03.012>.
- [28] Kim S, Yeo H. A flow-based vulnerability measure for the resilience of urban road network. *Procedia - Soc Behav Sci* 2016;218:13–23. <https://doi.org/10.1016/j.sbspro.2016.04.006>.
- [29] Sun D (Jian), Guan S. Measuring vulnerability of urban metro network from line operation perspective. *Transp Res A: Policy Pract* 2016;94:348–59. <https://doi.org/10.1016/j.tra.2016.09.024>.
- [30] Li T, Rong L, Yan K. Vulnerability analysis and critical area identification of public transport system: a case of high-speed rail and air transport coupling system in China. *Transp Res A: Policy Pract* 2019;127:55–70. <https://doi.org/10.1016/j.tra.2019.07.008>.
- [31] Murray-tuite P. A comparison of transportation network resilience under simulated system optimum and user equilibrium conditions. In: *Proceedings of the 2006 Winter Simulation Conference. IEEE*; 2006. <https://doi.org/10.1109/wsc.2006.323240>.
- [32] D'Lima M, Medda F. A new measure of resilience: an application to the london underground. *Transp Res A: Policy Pract* 2015;81:35–46. <https://doi.org/10.1016/j.tra.2015.05.017>.
- [33] Jing W, Xu X, Pu Y. Route redundancy-based network topology measure of metro networks. *J Adv Transp* 2019;2019:1–12. <https://doi.org/10.1155/2019/4576961>.
- [34] Xu X, Chen A, Xu G, Yang C, Lam WHK. Enhancing network resilience by adding redundancy to road networks. *Transp Res E: Logist Transp Rev* 2021;154:102448. <https://doi.org/10.1016/j.tre.2021.102448>.
- [35] Xu X, Chen A, Jansuwan S, Yang C, Ryu S. Transportation network redundancy: complementary measures and computational methods. *Transp Res B Methodol* 2018;114:68–85. <https://doi.org/10.1016/j.trb.2018.05.014>.
- [36] Zhang W, Wang N. Resilience-based risk mitigation for road networks. *Struct Saf* 2016;62:57–65. <https://doi.org/10.1016/j.strusafe.2016.06.003>.
- [37] Wu Y, Chen S. Resilience modeling and pre-hazard mitigation planning of transportation network to support post-earthquake emergency medical response. *Reliab Eng Syst Saf* 2023;230:108918. <https://doi.org/10.1016/j.res.2022.108918>.
- [38] Qu K, Fan X, Xu X, Hanasusanto GA, Chen A. Improving transportation network redundancy under uncertain disruptions via retrofitting critical components. *Transp Res B: Methodol* 2025;194:103174. <https://doi.org/10.1016/j.trb.2025.103174>.
- [39] Zhang T, Niu C, Nair DJ, Robson EN, Dixit V. Transportation resilience optimization from an economic perspective at the pre-event stage. *Sustain Anal Model* 2023;3:100027. <https://doi.org/10.1016/j.samod.2023.100027>.
- [40] Sang M, Ding Y, Bao M, Li S, Ye C, Fang Y. Resilience-based restoration strategy optimization for interdependent gas and power networks. *Appl Energy* 2021;302:117560. <https://doi.org/10.1016/j.apenergy.2021.117560>.
- [41] Dui H, Zheng X, Wu S. Resilience analysis of maritime transportation systems based on importance measures. *Reliab Eng Syst Saf* 2021;209:107461. <https://doi.org/10.1016/j.res.2021.107461>.
- [42] Sohounou PYR, Neves LAC. Assessing the effects of link-repair sequences on road network resilience. *Int J Crit Infrastruct Prot* 2021;34:100448. <https://doi.org/10.1016/j.ijcip.2021.100448>.
- [43] Zhang W, Wang N, Nicholson C. Resilience-based post-disaster recovery strategies for road-bridge networks. *Struct Infrastruct Eng* 2017;13:1404–13. <https://doi.org/10.1080/15732479.2016.1271813>.
- [44] Li Z, Jin C, Hu P, Wang C. Resilience-based transportation network recovery strategy during emergency recovery phase under uncertainty. *Reliab Eng Syst Saf* 2019;188:503–14. <https://doi.org/10.1016/j.res.2019.03.052>.
- [45] Zhao T, Zhang Y. Transportation infrastructure restoration optimization considering mobility and accessibility in resilience measures. *Transp Res C: Emerg Technol* 2020;117:102700. <https://doi.org/10.1016/j.trc.2020.102700>.
- [46] Yin H, Han B, Li D, Wang Y. Evaluating disruption in rail transit network: a case study of Beijing subway. *Procedia Eng* 2016;137:49–58. <https://doi.org/10.1016/j.proeng.2016.01.233>.
- [47] Dong S, Gao X, Mostafavi A, Gao J, Gangwal U. Characterizing resilience of flood-disrupted dynamic transportation network through the lens of link reliability and stability. *Reliab Eng Syst Saf* 2023;232:109071. <https://doi.org/10.1016/j.res.2022.109071>.
- [48] Miandoabchi E, Farahani RZ, Dullaert W, Szeto WY. Hybrid evolutionary metaheuristics for concurrent multi-objective design of urban road and public transit networks. *Netw Spat Econ* 2012;12:441–80. <https://doi.org/10.1007/s11067-011-9163-x>.
- [49] Karimi H, Ghadirifaraz B, Shetab Boushehri SN, Hosseiniinasab S-M, Rafiei N. Reducing traffic congestion and increasing sustainability in special urban areas through one-way traffic reconfiguration. *Transportation (Amst)* 2022;49:37–60. <https://doi.org/10.1007/s11116-020-10162-4>.
- [50] Gu Y, Fu X, Liu Z, Xu X, Chen A. Performance of transportation network under perturbations: reliability, vulnerability, and resilience. *Transp Res E: Logist Transp Rev* 2020;133:101809. <https://doi.org/10.1016/j.tre.2019.11.003>.
- [51] Barati H, Yazici A, Almotahari A. A methodology for ranking of critical links in transportation networks based on criticality score distributions. *Reliab Eng Syst Saf* 2024;251:110332. <https://doi.org/10.1016/j.res.2024.110332>.
- [52] Fan Y, Ding J, Liu H, Wang Y, Long J. Large-scale multimodal transportation network models and algorithms-part I: the combined mode split and traffic assignment problem. *Transp Res E: Logist Transp Rev* 2022;164:102832. <https://doi.org/10.1016/j.tre.2022.102832>.
- [53] Hamilton AL, Reed PM, Gupta RS, Zeff HB, Charaklis GW. Resilient water infrastructure partnerships in institutionally complex systems face challenging supply and financial risk tradeoffs. *Nat Commun* 2024;15:7354. <https://doi.org/10.1038/s41467-024-51660-8>.
- [54] Nickdoost N, Jalloul H, Choi J, Smith D. Identification and prioritization of multidimensional resilience factors for incorporation in coastal state transportation infrastructure planning. *Nat Hazards* 2024;120:1603–63. <https://doi.org/10.1007/s11069-023-06251-z>.
- [55] Wang J, Huang J, Yang H, Levinson D. Resilience and recovery of public transport use during COVID-19. *NPJ Urban Sustain* 2022;2. <https://doi.org/10.1038/s42949-022-00061-1>.
- [56] Ma F, Ao Y, Wang X, He H, Liu Q, Yang D, et al. Assessing and enhancing urban road network resilience under rainstorm waterlogging disasters. *Transp Res D: Transp Environ* 2023;123:103928. <https://doi.org/10.1016/j.trd.2023.103928>.
- [57] Wei M, Xu J. Assessing road network resilience in disaster areas from a complex network perspective: a real-life case study from China. *Int J Disaster Risk Reduct* 2024;100:104167. <https://doi.org/10.1016/j.ijdrr.2023.104167>.
- [58] Xu F, Fang D, Chen B, Wang H. Resilience assessment of subway system to waterlogging disaster. *Sustain Cities Soc* 2024;113:105710. <https://doi.org/10.1016/j.scs.2024.105710>.
- [59] Wang L, Zhang S, Szűcs G, Wang Y. Identifying the critical nodes in multi-modal transportation network with a traffic demand-based computational method. *Reliab Eng Syst Saf* 2024;244:109956. <https://doi.org/10.1016/j.res.2024.109956>.
- [60] Chen X, Ma S, Chen L, Yang L. Resilience measurement and analysis of intercity public transportation network. *Transp Res D: Transp Environ* 2024;131:104202. <https://doi.org/10.1016/j.trd.2024.104202>.
- [61] Ma W, Lin S, Ci Y, Li R. Resilience evaluation and improvement of post-disaster multimodal transportation networks. *Transp Res A: Policy Pract* 2024;189:104243. <https://doi.org/10.1016/j.tra.2024.104243>.
- [62] Xiao Y, Li Y, Liu W, Wang Z, Chen J, Wang W. Resilience assessment of high-speed railway networks from the spatio-temporal perspective: a case study in Jiangsu Province, China. *Reliab Eng Syst Saf* 2025;259:110900. <https://doi.org/10.1016/j.res.2025.110900>.
- [63] Salvo G, Karakikes I, Papaioannou G, Polydoropoulou A, Sanfilippo L, Brignone A. Enhancing urban resilience: managing flood-induced disruptions in road networks. *Transp Res Interdiscip Perspect* 2025;31:101383. <https://doi.org/10.1016/j.trip.2025.101383>.
- [64] Hamedmoghadam H, Jalili M, Vu HL, Stone L. Percolation of heterogeneous flows uncovers the bottlenecks of infrastructure networks. *Nat Commun* 2021;12:1254. <https://doi.org/10.1038/s41467-021-21483-y>.
- [65] Adjetej-Bahun K, Birregah B, Châtelet E, Planchet J-L. A model to quantify the resilience of mass railway transportation systems. *Reliab Eng Syst Saf* 2016;153:1–14. <https://doi.org/10.1016/j.res.2016.03.015>.
- [66] Argyroudis SA, Mitoulis SA, Winter MG, Kaynia AM. Fragility of transport assets exposed to multiple hazards: state-of-the-art review toward infrastructural resilience. *Reliab Eng Syst Saf* 2019;191:106567. <https://doi.org/10.1016/j.res.2019.106567>.
- [67] Artimo O, Grassia M, De Domenico M, Gleeson JP, Makse HA, Mangioni G, et al. Robustness and resilience of complex networks. *Nat Rev Phys* 2024;6:114–31. <https://doi.org/10.1038/s42254-023-00676-y>.

- [68] Li W, Wang J, Fan R, Zhang Y, Guo Q, Siddique C, et al. Short-term traffic state prediction from latent structures: accuracy vs. efficiency. *Transp Res C Emerg Technol* 2020;111:72–90. <https://doi.org/10.1016/j.trc.2019.12.007>.
- [69] Huang K, Dai L, Yao M, Fan Y, Kong X. Modelling dependence between traffic noise and traffic flow through an entropy-copula method. *J Environ Inform* 2017;29(2): 134–51. <https://doi.org/10.3808/jei.201500302>.
- [70] Florian M, Mahut M, Tremblay N. Application of a simulation-based dynamic traffic assignment model. *Eur J Oper Res* 2008;189:1381–92. <https://doi.org/10.1016/j.ejor.2006.07.054>.
- [71] Laña I, Lobo JL, Capecci E, Del Ser J, Kasabov N. Adaptive long-term traffic state estimation with evolving spiking neural networks. *Transp Res C Emerg Technol* 2019;101:126–44. <https://doi.org/10.1016/j.trc.2019.02.011>.
- [72] Khattak A, Wang X, Zhang H. Are incident durations and secondary incidents interdependent? *Transp Res Rec: J Transp. Res Board* 2009;2099:39–49. <https://doi.org/10.3141/2099-05>.
- [73] Li R, Pereira FC, Ben-Akiva ME. Overview of traffic incident duration analysis and prediction. *Eur Transp Res Rev* 2018;10:22. <https://doi.org/10.1186/s12544-018-0300-1>.
- [74] Engsig M, Tejedor A, Moreno Y, et al. DomRank centrality reveals structural fragility of complex networks via node dominance. *Nat Commun* 2024;15:56. <https://doi.org/10.1038/s41467-023-44257-0>.